"Sustainable Design"

By

D. AMARANATHA REDDY
ASSOCIATTE PROFESSOR
DÉPARTEMENT OF SCIENCES
Indian Institute of Information Technology Design and Manufacturing, Kurnool, Andhra Pradesh, India

Sustainable Development Goal & Importance of SD

The World Commission on Environment and Development defines sustainable development as:

"Development that meets the needs of the present without compromising the needs of future generations to meet their own needs."



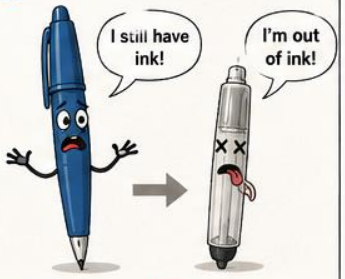
Sustainable Design

1 Mobile Charger Syndrome



One charger for many devices.
Smart design, less waste.

2 The Pen Mystery



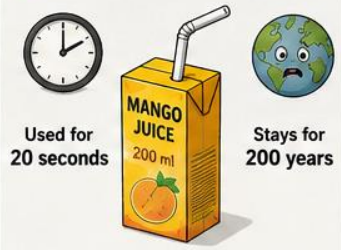
We throw away 95% good material because 5% stopped working!

3 Shampoo Bottle Gym



We transport mostly water.
Why not shampoo tablets?

4 The Mango Juice Straw



Who designed this relationship?

5 Birthday Balloon Logic



Did the balloon enjoy the birthday more than we did?

6 Wedding Water Bottle



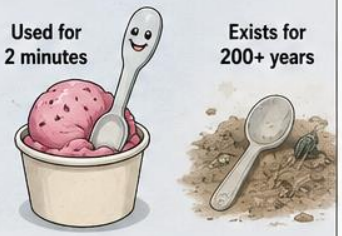
Was the bottle designed for drinking water or decorating dustbins?

7 Engineering Student Notebook



Poor utilization is also poor design.

8 Plastic Spoon Ice Cream



Is the spoon more committed to Earth than we are?

9 Remote Control Collection



Why does every appliance need its own PhD in button design?

10 The Cable Drawer



Nobody knows.
Yet nobody throws it away.

11 Pizza Box Problem



Who designed a package bigger than the food?

12 Toothpaste Tube



A product that cannot be recycled was never designed for circularity.

13 Coffee Cup Challenge



So... the cup is pretending to be environmentally friendly?

14 Fast Fashion



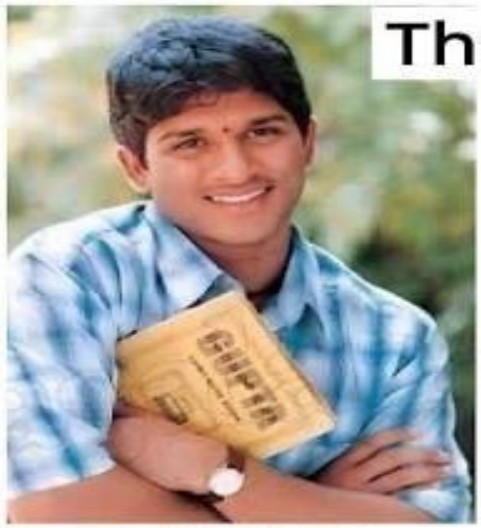
Did you buy clothes or adopt future textile waste?

15 Amazon Packaging



Was the package protecting the cable, or the cable protecting the package?

Sustainable Development



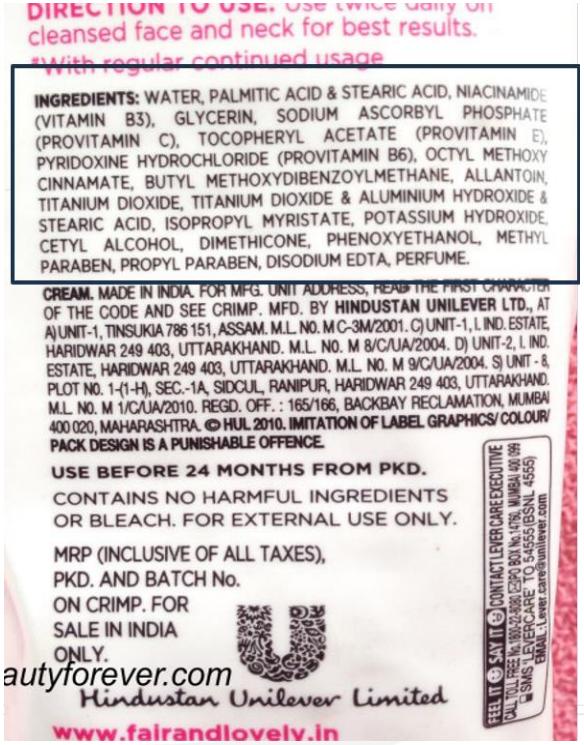












Component	Approximate Quantity
Cream (product)	**60 g**
Plastic tube body	**5–7 g**
Plastic cap	**2–3 g**
Total plastic packaging	**7–10 g**

Sustainable Development

Classroom Calculation

Suppose:

- 80 students
- Each uses **2 cream tubes/months**

That's

80 × 24 = 1920 plastic tubes/year

Now imagine

- 1000 students in institute

1000 × 1920 = 1920000 plastic tubes every year

Last Drop Engineering

"Why do companies make tubes impossible to empty?

:"To sell another one!"

"As engineers, our job is exactly the opposite.

"Good design: ✅ Empty 99% of the product

Bad design: ❌ Leave 15% inside forever.

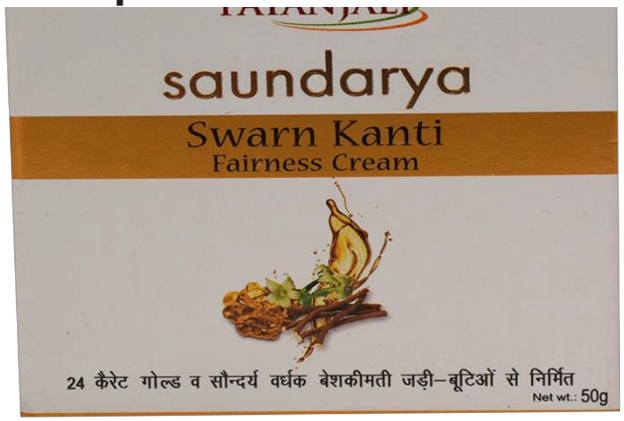
SDG	Connection
SDG 12	Responsible Consumption
SDG 13	Climate Action
SDG 11	Sustainable Cities
SDG 9	Innovation
SDG 3	Health & Well-being

Looking beautiful for one month shouldn't create waste that lasts for centuries. Sustainability begins when we rethink what happens after the product is finished."

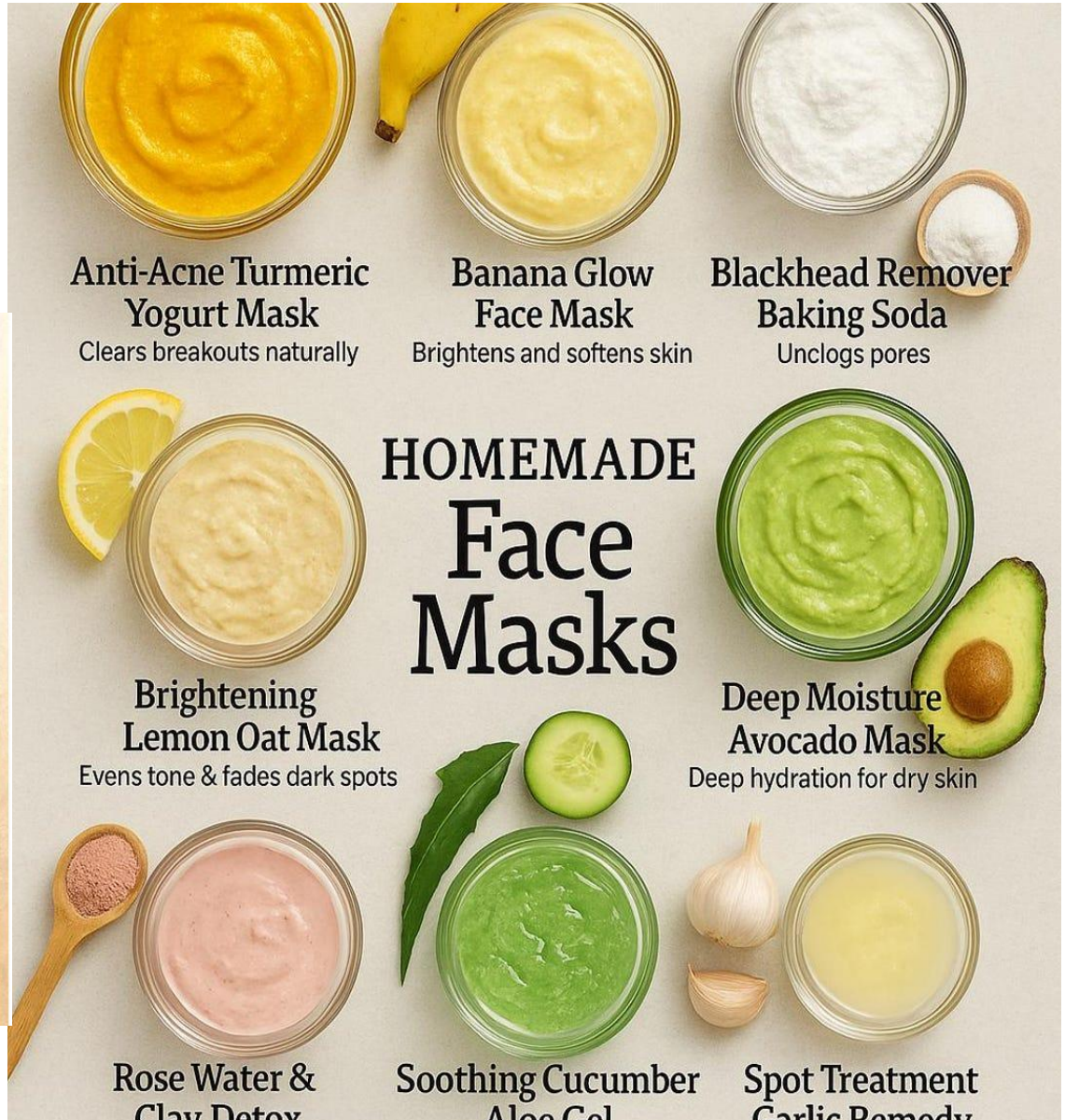
BUY MORE, WASTE MORE!



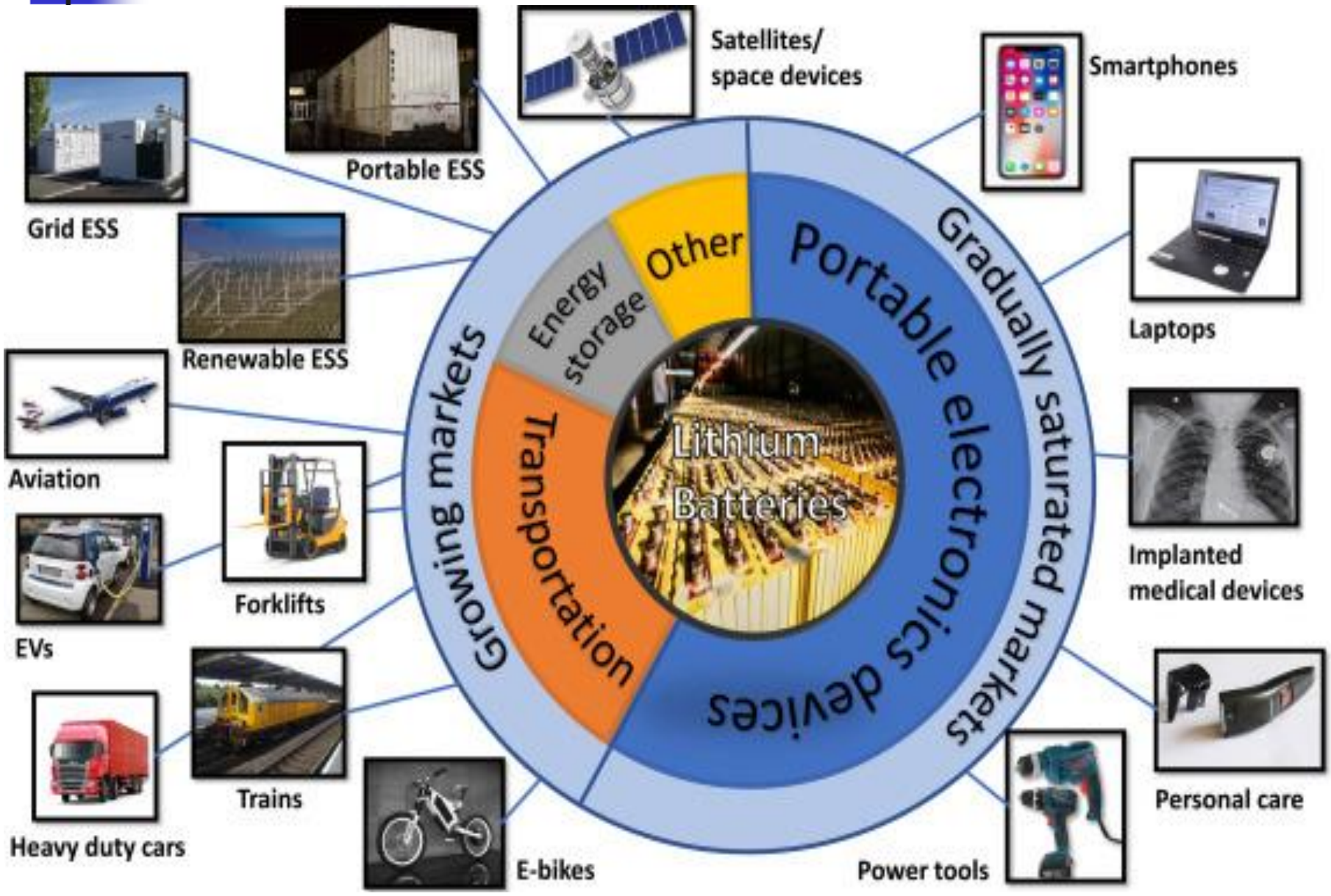
Fair Skin or a Fair Future?
Think Before you use





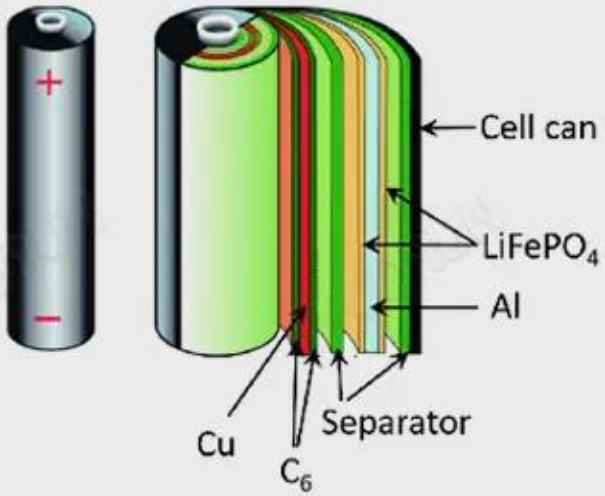


Batteries


Lithium Batteries
Growing markets
Gradually saturated markets
Portable electronics devices
Transportation
Energy storage
Other
Portable ESS
Satellites/ space devices
Smartphones
Grid ESS
Renewable ESS
Laptops
Aviation
Forklifts
Implanted medical devices
EVs
Trains
Personal care
Heavy duty cars
E-bikes
Power tools

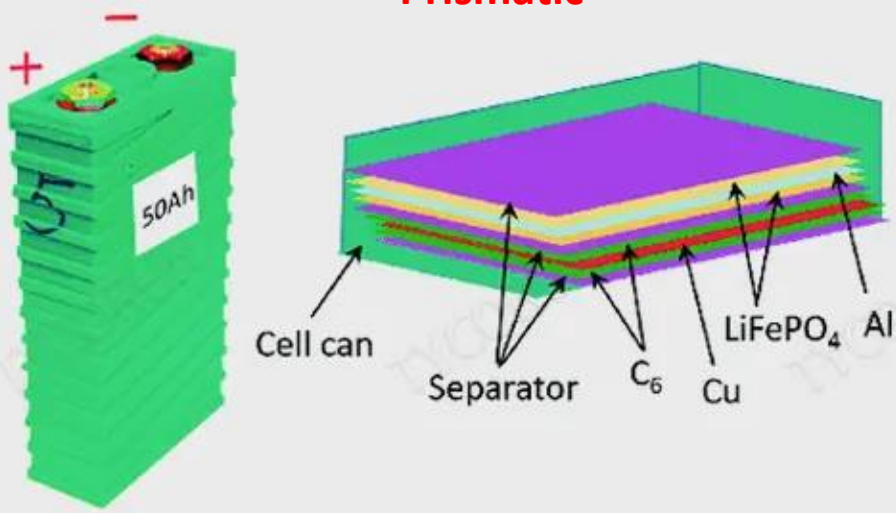
Introduction of different lithium battery ingredients - characteristics and applications

Cylindrical



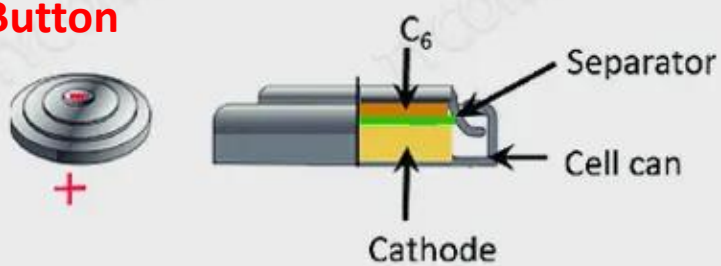
(a)

Prismatic



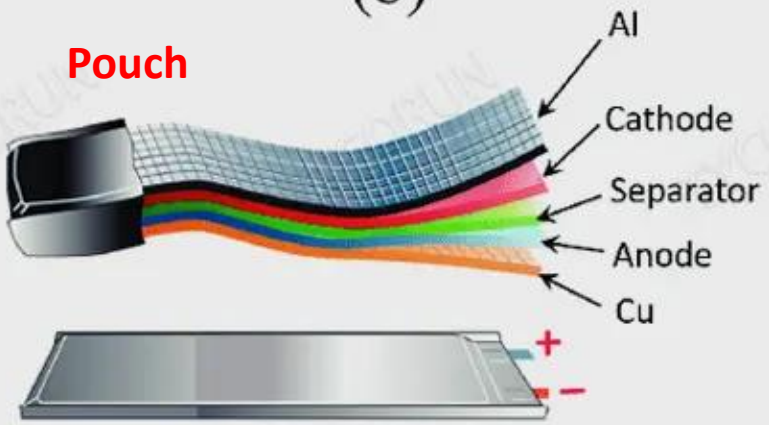
(b)

Button



(c)

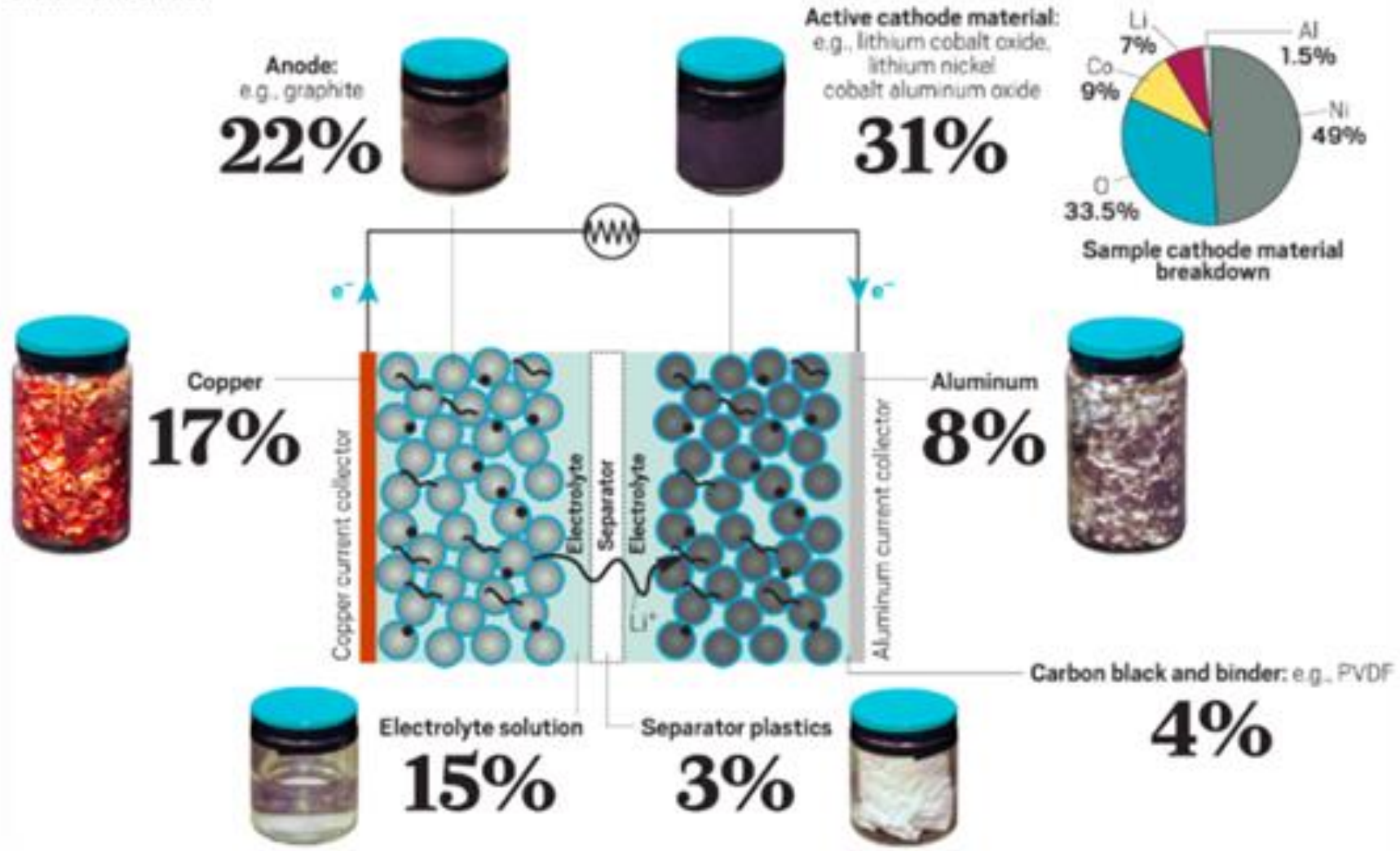
Pouch



(d)

Inside a Li-ion battery

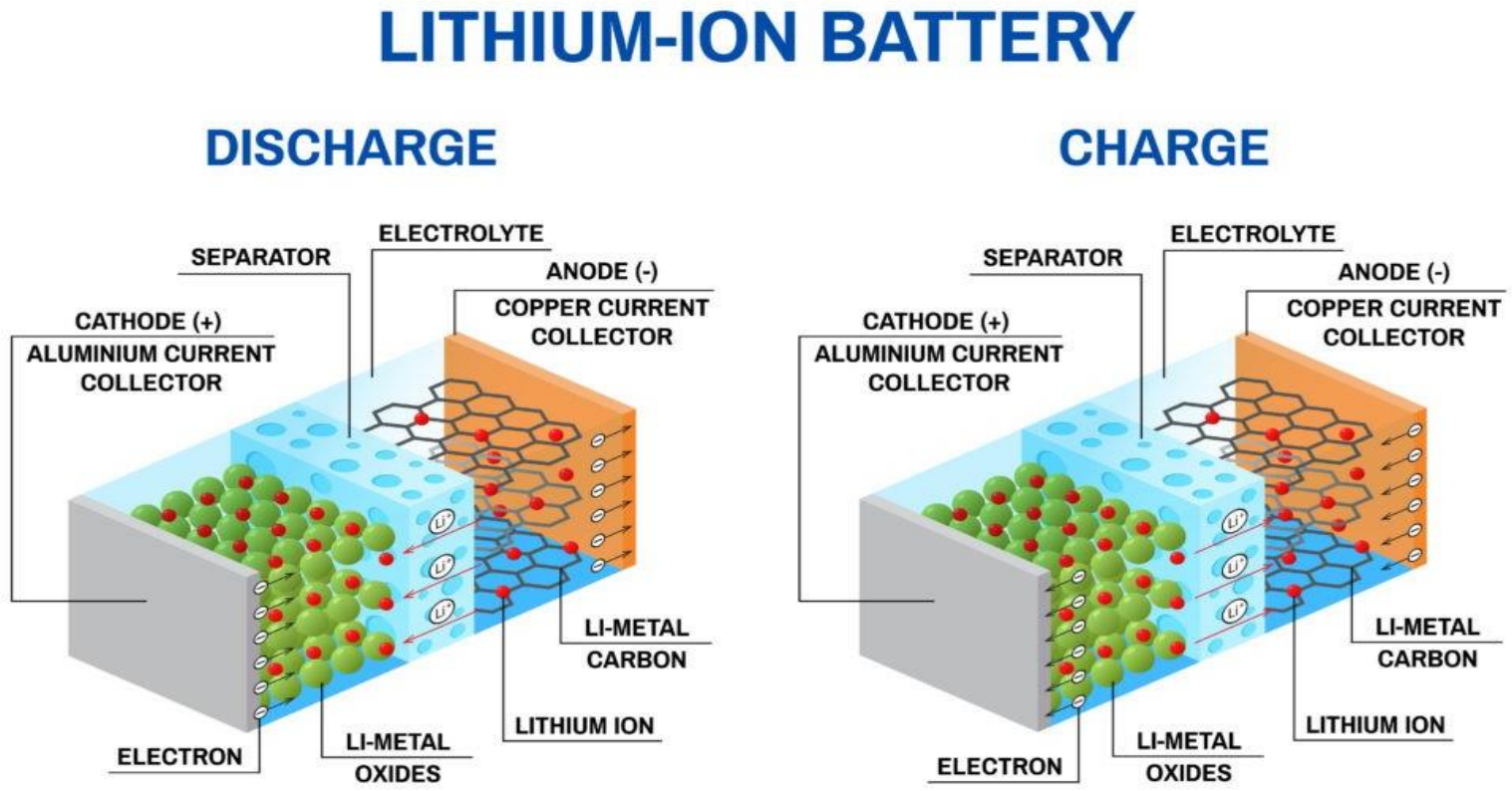
All the components of a Li-ion battery have value and can be recovered and reused. Currently, most recyclers recover just the metals. The pie chart describes a cathode material known as NCA, which is made of lithium nickel cobalt aluminum oxide.



Credit: Mitch Jacoby/C&EN

Source: Argonne National Laboratory.

Working Principle



Process

- The battery powers a device.
- Lithium ions leave the anode.
- They move through the electrolyte to the cathode.
- Electrons flow through the external circuit, producing electric current.
- At the cathode, lithium ions and electrons recombine.

Process

- An external charger supplies electrical energy.
- Lithium ions leave the cathode.
- The ions travel through the electrolyte to the anode.
- Electrons travel through the external circuit to the anode.
- Lithium ions combine with electrons and are stored between graphite layers.
- Energy is stored as chemical energy.

Recycling and reusability is the unique choice to solve these issues

- Reduce deposition of toxic waste sent to landfill
- Preventing pollution by removal of hazardous wastes
- Supply precious metals to the circular economy supply chain
- Cutdown global warming by the mining for precious metals
- Reduce the risk of fire & explosion
- Reduce the human and environmental impact
- Recover non-renewable materials.
- Conserving Natural Resources



Benefits to India

- Asset to Indian energy storage manufacturing sector
- Making the country self-reliant
- Contributes to the economy
- Reduce the cost of new batteries
- Cutdown the importing raw materials and batteries
- Secondary applications
- Producing New Products and New Jobs

- All the components of a Li-ion battery can be recycled
- Cathode and anode current collectors such as copper, and aluminum.
- Casings steel, and plastics
- Pure metals like Li, Co, Ni, Fe, Mn, etc
- Organic compounds such as electrolyte
- Anode material like graphite

Sustainable Design Syllabus
Module-I

- *Role and Importance of Sustainable Development*
- *Sustainable Development Goals*
- *Three pillars of sustainability*
- *Sustainable Product Design*
- *Principles of sustainable Design Process*
- *Selection of Raw Materials*
- *Sustainable Approach followed in manufacturing*
- *Techniques and tools for sustainability measurement*

Sustainable Design

Module-II

Sustainable Hydrogen Energy: Fuel Cells Design

- ***Its merit as fuel Importance***
- ***Hydrogen Fuel production technologies from fossil fuels,***
- ***Sustainable methods for Hydrogen Fuel Production***
- ***Hydrogen Fuel storage technologies working process and design process.***
- ***Design process of Fuel cells and their application in power and transportation (Zero emission vehicles).***

Sustainable Design Syllabus

Module-III

Green Energy Technologies for Design & Manufacturing

- *Non-Renewable Energy Resources: Present scenario and future prospects*
- *Types of Green energy,*
- *Basics of green energy production*
- *Energy Harvesting Techniques*
- *Basic concepts and working principals of Solar Energy utilization*
- *Solar Energy Based Devices*

Sustainable Design
Module-III

Energy Harvesting Techniques

- *Solar Cells*
- *Solar collectors and types: flat plate, concentrating solar collectors, advanced collectors and solar concentrators, selective coatings,*
- *Solar water heating,*
- *Solar cooking, Solar drying, solar distillation and solar refrigeration, Active and passive heating and cooling of buildings,*
- *solar thermal power generation, lighting systems, solar PV pumps, solar energy storage options*
- *Batteries; Types, Design and utilization*
- *Super Capacitors alternative to batteries*

Sustainable Design

Module-IV

- *Types and Sources of solid and hazardous wastes*
- *waste processing technologies-contribution of fossil fuels emission to climate change Wastewater treatment technologies*
- *Global warming: Carbon capture and conversion devices design and manufacturing process*

Sustainable Design Syllabus
Module-I

- *Role and Importance of Sustainable Development*
- *Sustainable Development Goals*
- *Three pillars of sustainability*
- *Sustainable Product Design*
- *Principles of sustainable Design Process*
- *Selection of Raw Materials*
- *Sustainable Approach followed in manufacturing*
- *Techniques and tools for sustainability measurement*

Sustainable Development Goal & Importance of SD

The World Commission on Environment and Development defines sustainable development as:

"Development that meets the needs of the present without compromising the needs of future generations to meet their own needs."



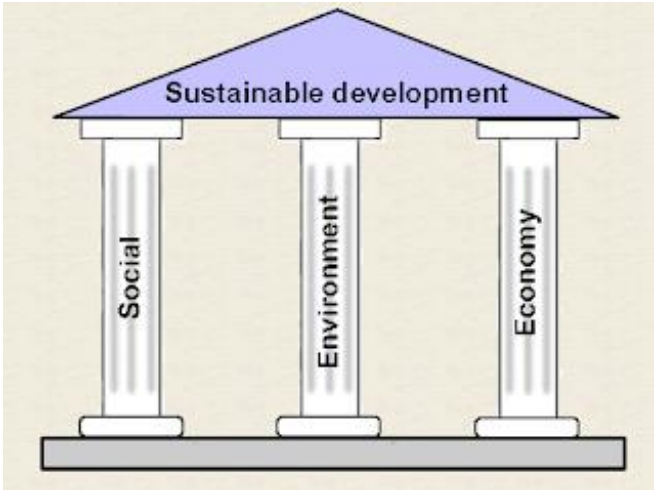
Objectives of sustainable development

- Economic growth
- Environmental protection
- Social inclusion



Importance of Sustainable Development

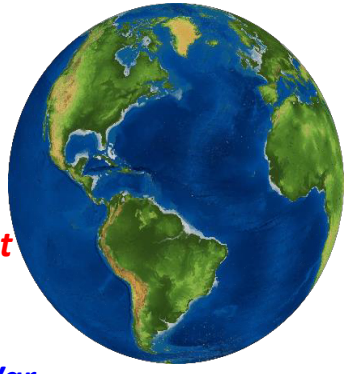
1. Provides Essential Human Needs
2. Agricultural Requirement
3. Manage Climate Change
4. Financial Stability
5. Sustain Biodiversity



Sustainable Energy Technologies

Humanity's Top Ten Problems for next 50 Years

1. *Energy*
2. *Water*
3. *Food*
4. *Environment*
5. *Poverty*
6. *Terrorism & War*
7. *Disease*
8. *Education*
9. *Democracy*
10. *Population*



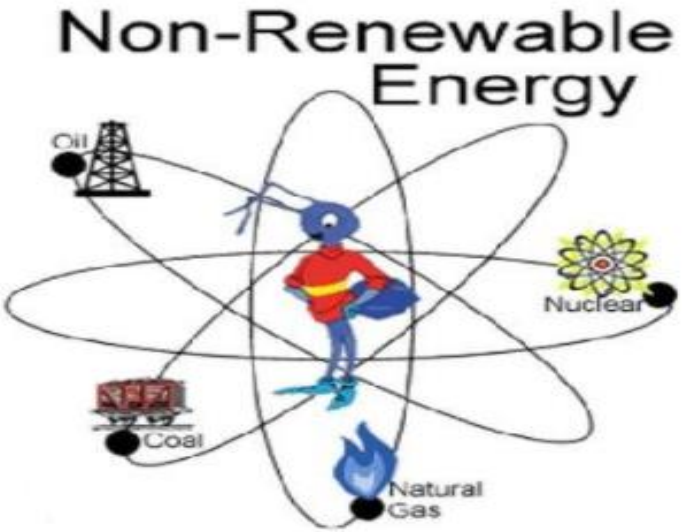
Survey done by Rice University Students. USA

What are non-renewable energy sources?

- Non-renewable energy resources are finite resources that come out of the earth as liquids, gases, and solids.
- These sources are called nonrenewable because they do not form or restore at a sufficient rate for sustainable economic extraction in meaningful human time-frames.
- Their formation typically takes millions of years. These resources would, therefore, run out one day if they are continually extracted and used by us.

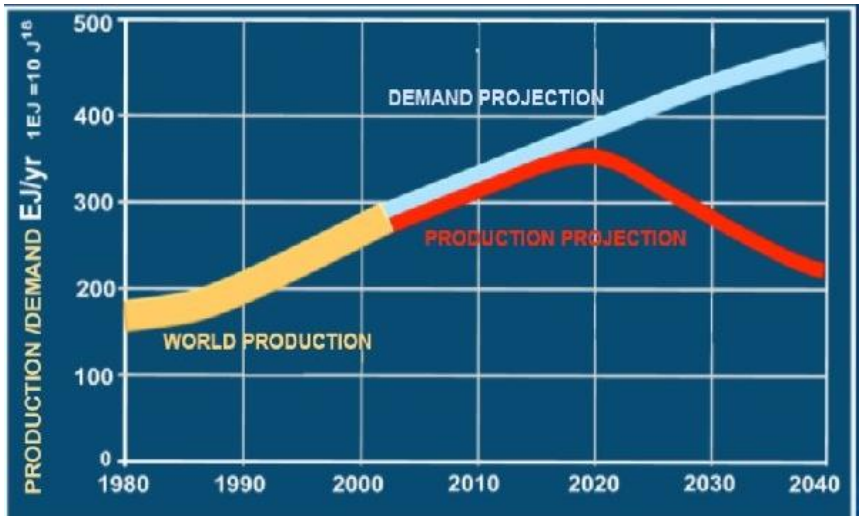
Examples:

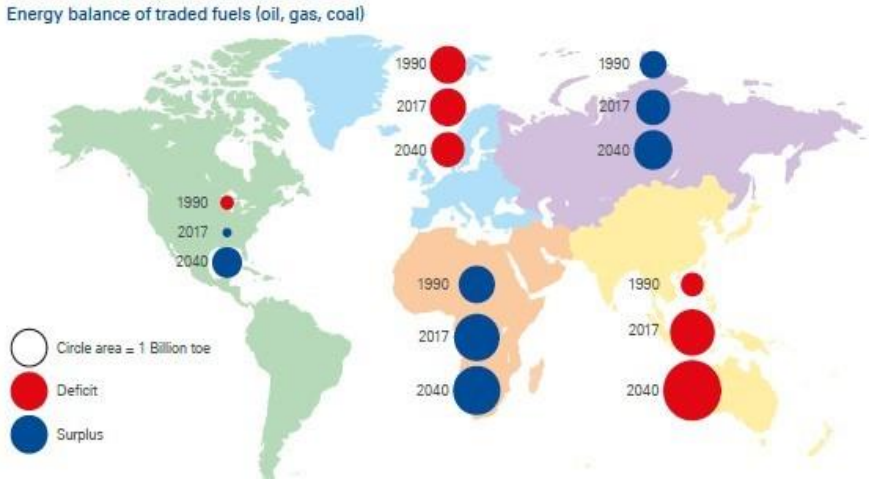
- Coal
- Crude Oil
- Natural Gas



- Most of the non-renewable energy sources are fossil fuels i.e. Coal, Crude oil and Natural gas.

Fossil Fuel Production/Demand (Petroleum and Natural gas)

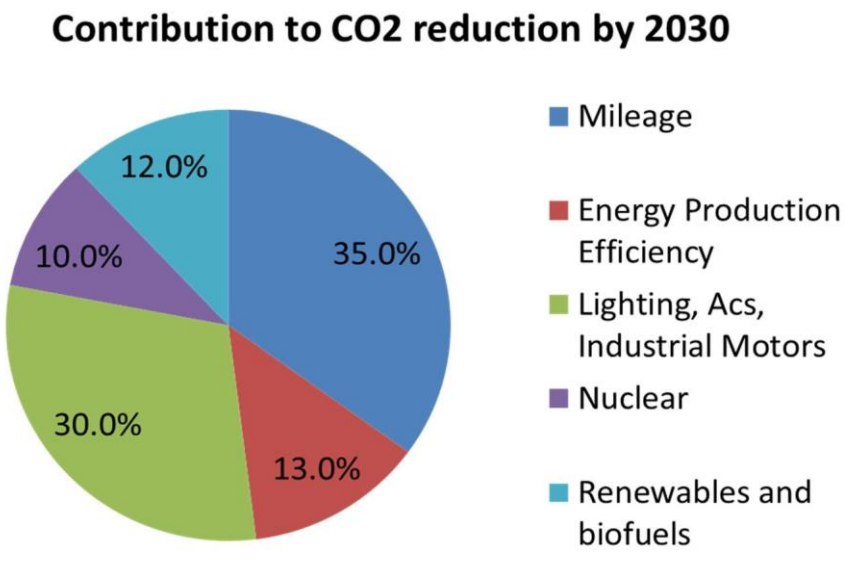




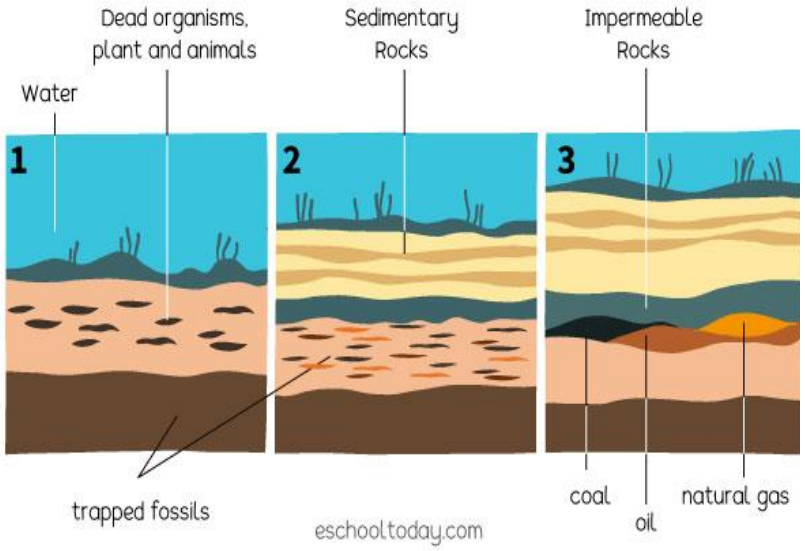
India Imports

Projected CO_2 Reductions from Various Sources

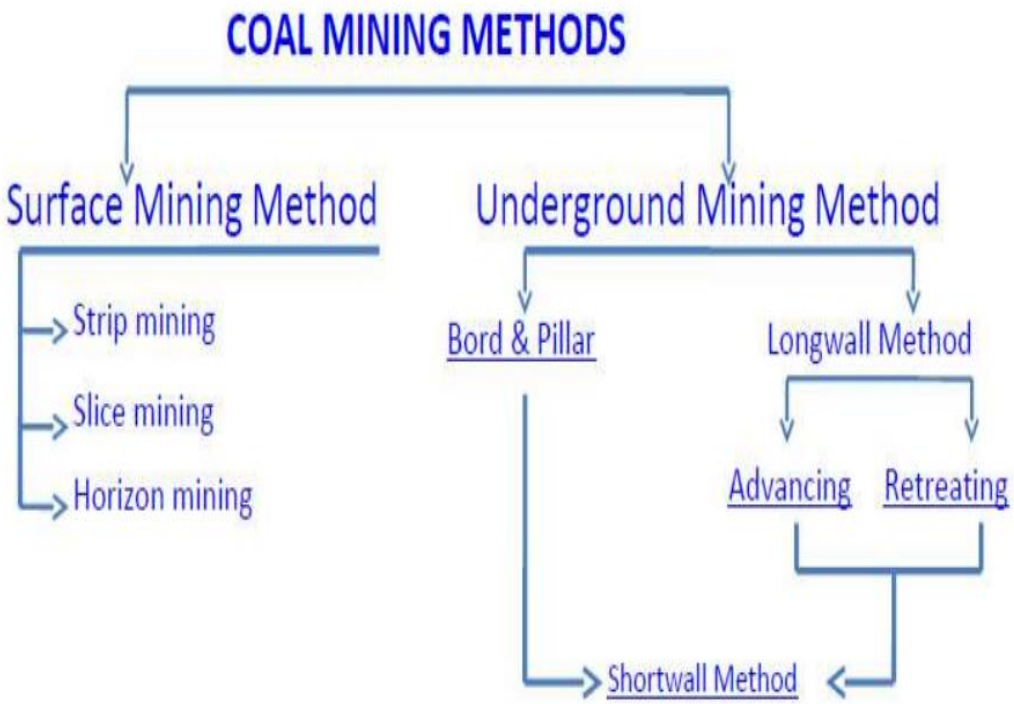
$480B (2025)



How are non-renewable resources formed?



Coal Extraction:



Coal Demerits

The mining of coal creates dust pollution

Burning of coal releases pollutants (Carbon and sulphur)

- Sulphur leads to acid mine drainage and acid deposition

Also mercury is released as a result of the burning of coal.

The increased amount of CO_2 release to the atmosphere causes severe environmental problems and ultimately results into global warming .

(ii) Crude Oil

Steps:

- **Crude oil is heated** until it vaporizes.
- **Vapour enters** the bottom of the distillation tower.
- **Vapour rises** through the tower.
- **Temperature decreases** with increasing height.
- **Hydrocarbons condense** at different levels based on their boiling points.
- **Fractions are collected**, each containing similar-sized hydrocarbon molecules.
- **Boiling point differences** enable separation into multiple products.
- **Further refining** removes impurities and improves fuel quality

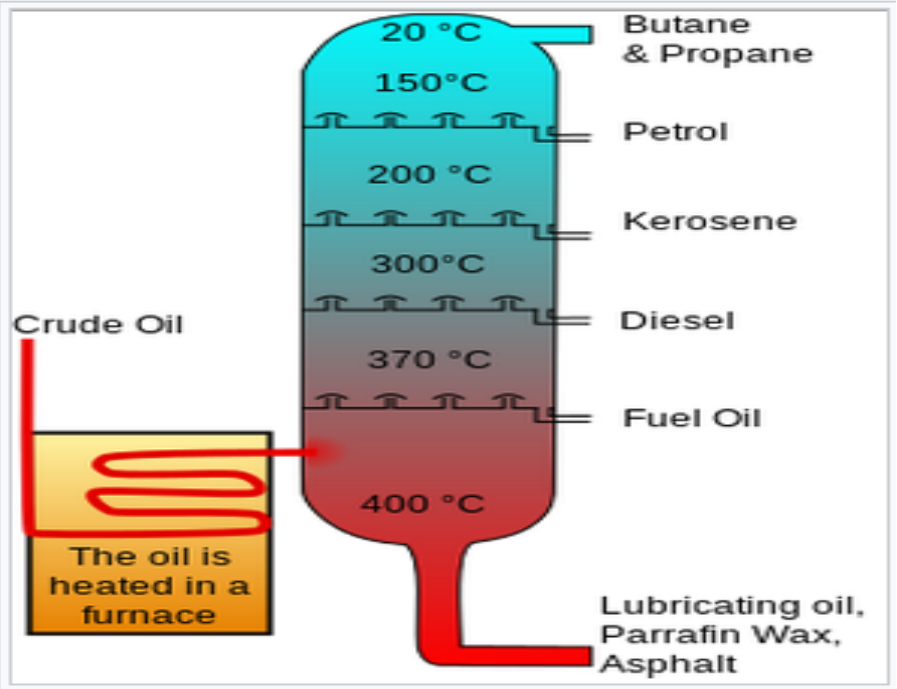
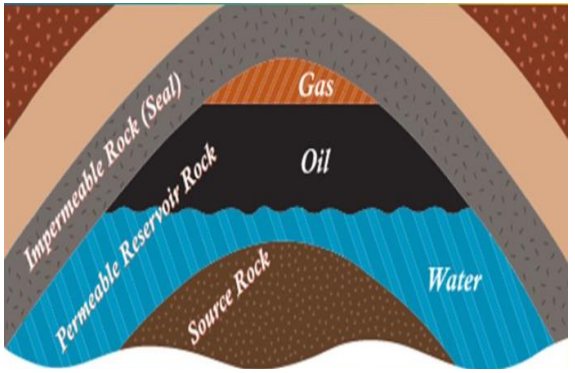


Diagram of a fractional distillation tower, showing where the different fractions will condense.[1] Note that the temperature is higher at the bottom, so the longer carbon chains will fall out at the bottom, the shorter carbon chains will go up the column until they hit a temperature at which they become liquid.

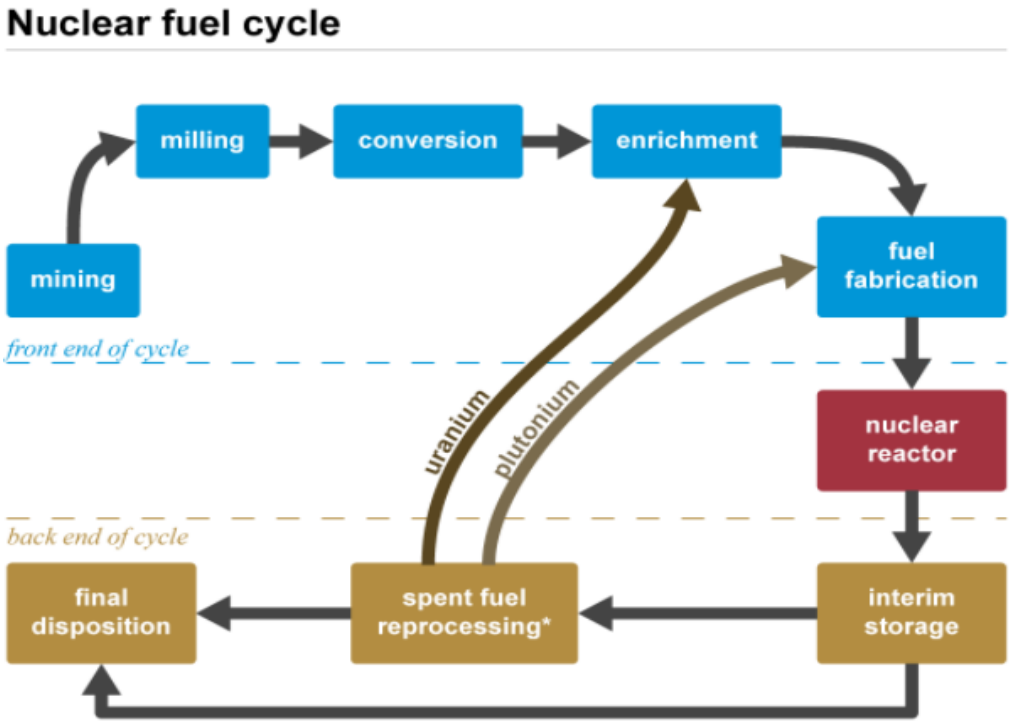
Schematic of fractional distillation



(iv) Nuclear Fuels

- **Nuclear fuel** powers nuclear reactors.
- **Main fuels:** Uranium-235 (U-235) and Plutonium-239 (Pu-239).
- **Fission reaction** releases a large amount of heat.
- **Nuclear fuel cycle:** Mining → Refining → Fuel fabrication → Reactor use.
- **Uranium ore** is mined, purified, and converted into reactor fuel.
- **Heat from the reactor** produces steam.
- **Steam drives turbines** to generate electricity.





Advantages & Disadvantages of non-renewable energy sources:

- There are a few advantages of non renewable energy sources.
- Fossil fuels, such as coal, crude oil and natural gas are a valuable source of energy with high energy content.
- They can also be stored, piped, or shipped anywhere in the world. Unlike many renewable energy sources, fossil fuels are relatively less expensive to produce.
- The probable reason why fossil fuels are in higher demand is that it tends to cost less.
- Fuels created from non-renewable resources are still the primary source of all the power generated in the world because of their affordability and high energy content.

Enforcing Regulatory Policies for Internal Combustion Engines

Internal Combustion Ban Opens Up New Opportunities for ZEV's

EU

Agree to 30% cut of truck & bus CO_2 emissions by 2030
(Interim target of a 15% reduction by 2025, based on 2019 levels)

※ Car by 37.5% reduction, Van by 31% reduction (by 2030 compare to 2021)

NORWAY

Combustion Engine Vehicle's Sales Ban from 2025

CHINA

Restriction of newly established company with only conventionally fueled vehicles, including hybrids and plug-in hybrids from 2019 1/10.

GERMANY

German cities ban older diesel cars
(Frankfurt, Darmstadt, Berlin, Munich, Essen, Mainz, Hamburg etc.)

- Darmstadt (The state of Hesse)
  : To ban Euro 5 standard from mid-2019
- Frankfurt
  : Prohibited from entering the city center
  Euro 4 from Feb. 2019, Euro 5 from Sep.2019

UK / FRANCE

Diesel Engine Vehicle's Sales Ban from 2040

KOREA

Plan to ban Diesel Engine Vehicles from 2030



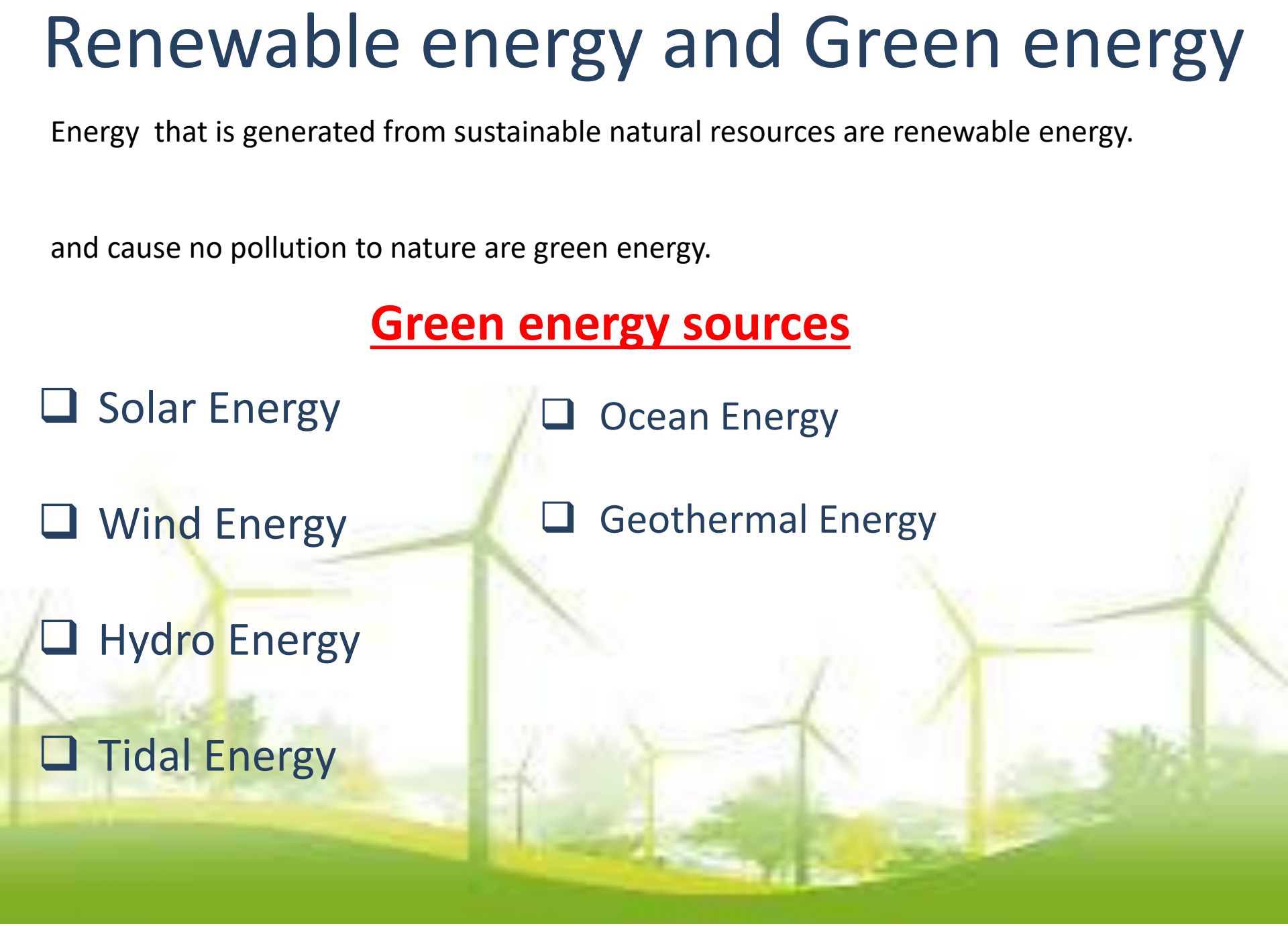
Renewable energy and Green energy

Energy that is generated from sustainable natural resources are renewable energy.

and cause no pollution to nature are green energy.

Green energy sources

- Solar Energy
- Wind Energy
- Hydro Energy
- Tidal Energy
- Ocean Energy
- Geothermal Energy

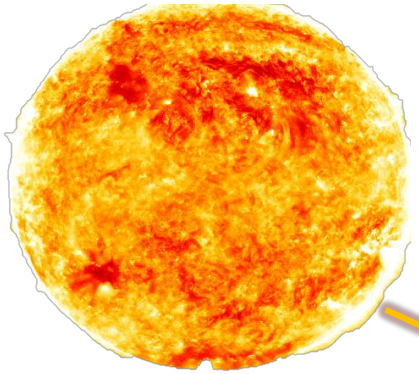


Importance of Solar Energy

Solar energy available on the surface of the landscape ca. 1x10^{15} W= 1000 TW

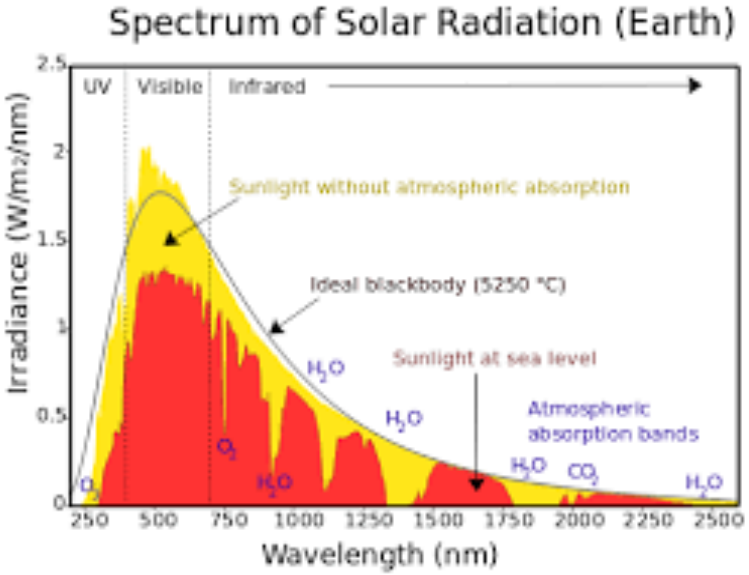
Total energy consumption by humankind (in 2004) ca. 1.3 x 1015 W = 13 TW

If we utilize 1% of the solar energy supplied it can possible to overcome present energy demand

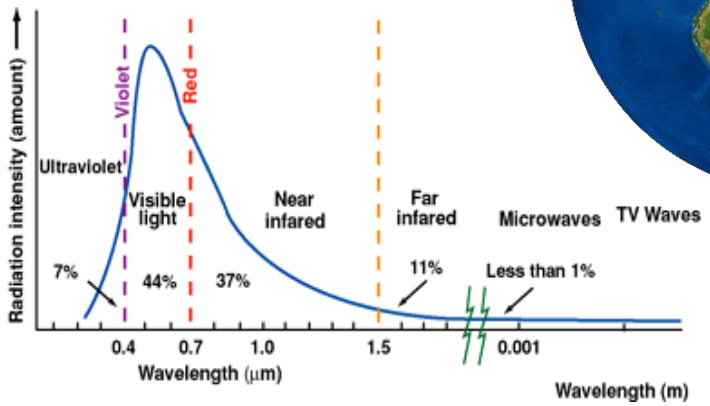


Supplied to the total surface (173 000 TW)

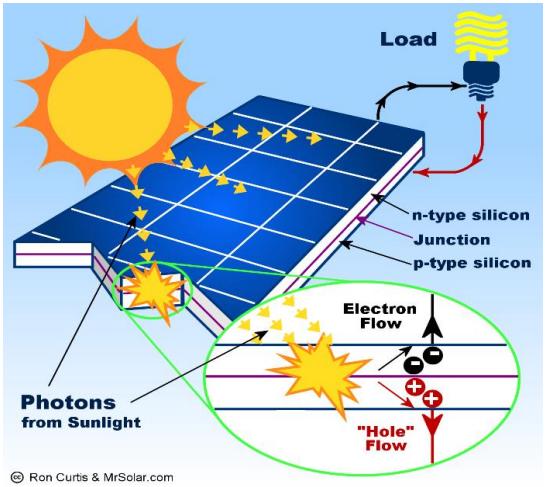
Equal to 350 nuclear power plants production







Possible ways of Solar Energy Conversion



- Solar cells
- 22 % max efficiency

Solar

to

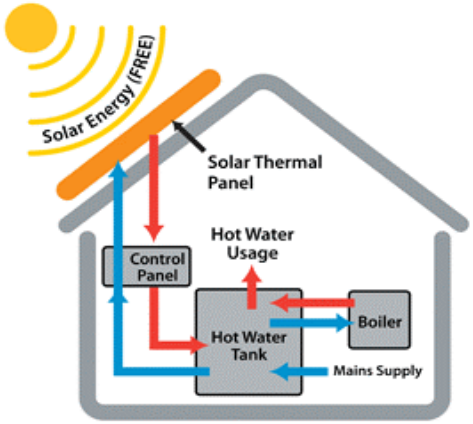
Electricity

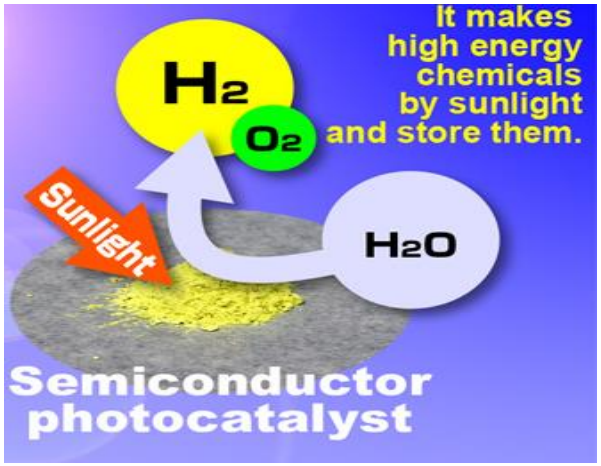
- Solar water heater
- Solar drying
- Solar cooking
- Solar distillation
- Over 30 % max efficiency

Solar

to

Thermal



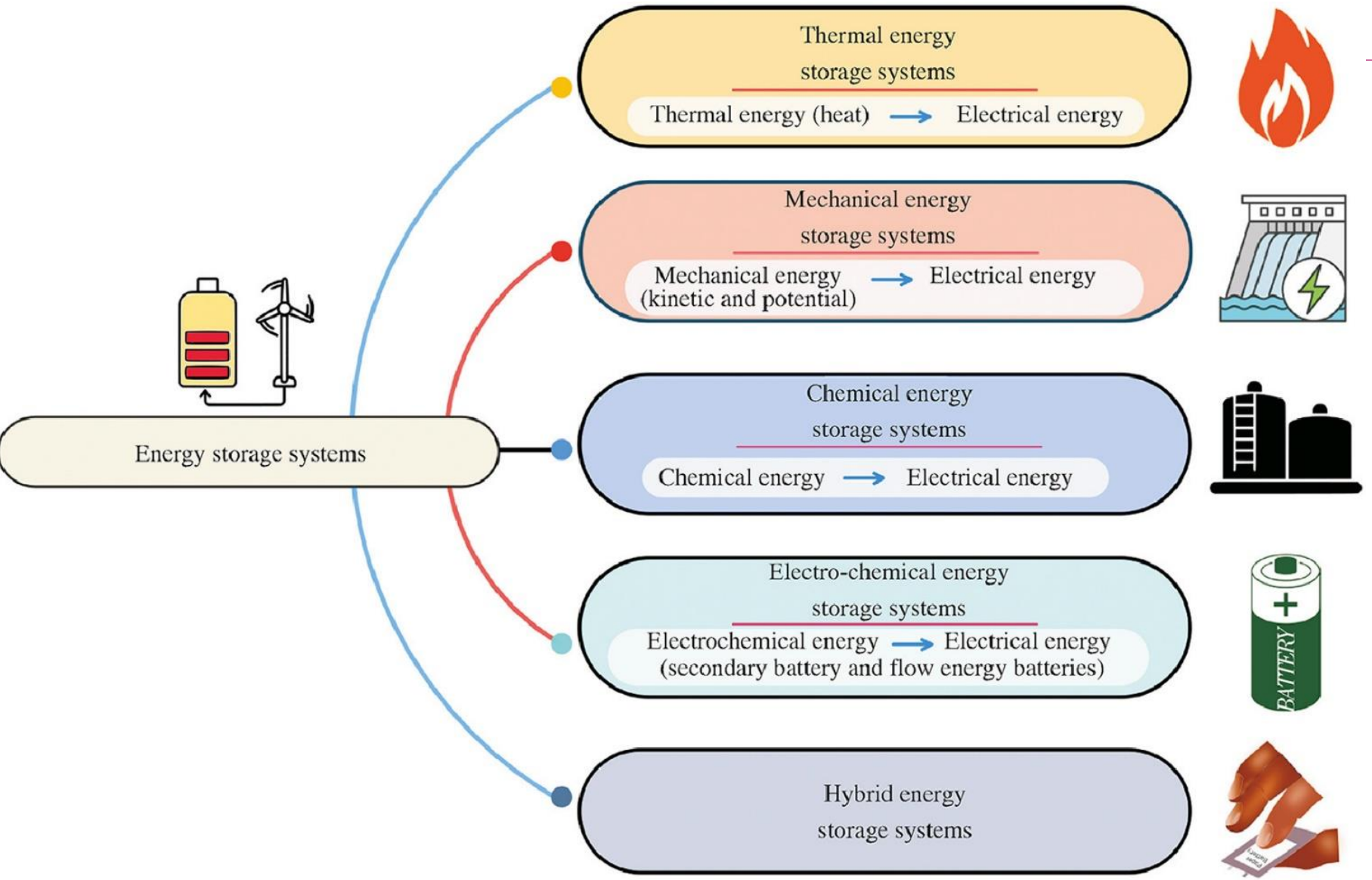


- Artificial Photosynthesis
- Solar water splitting

Solar

to

Chemical Fuels

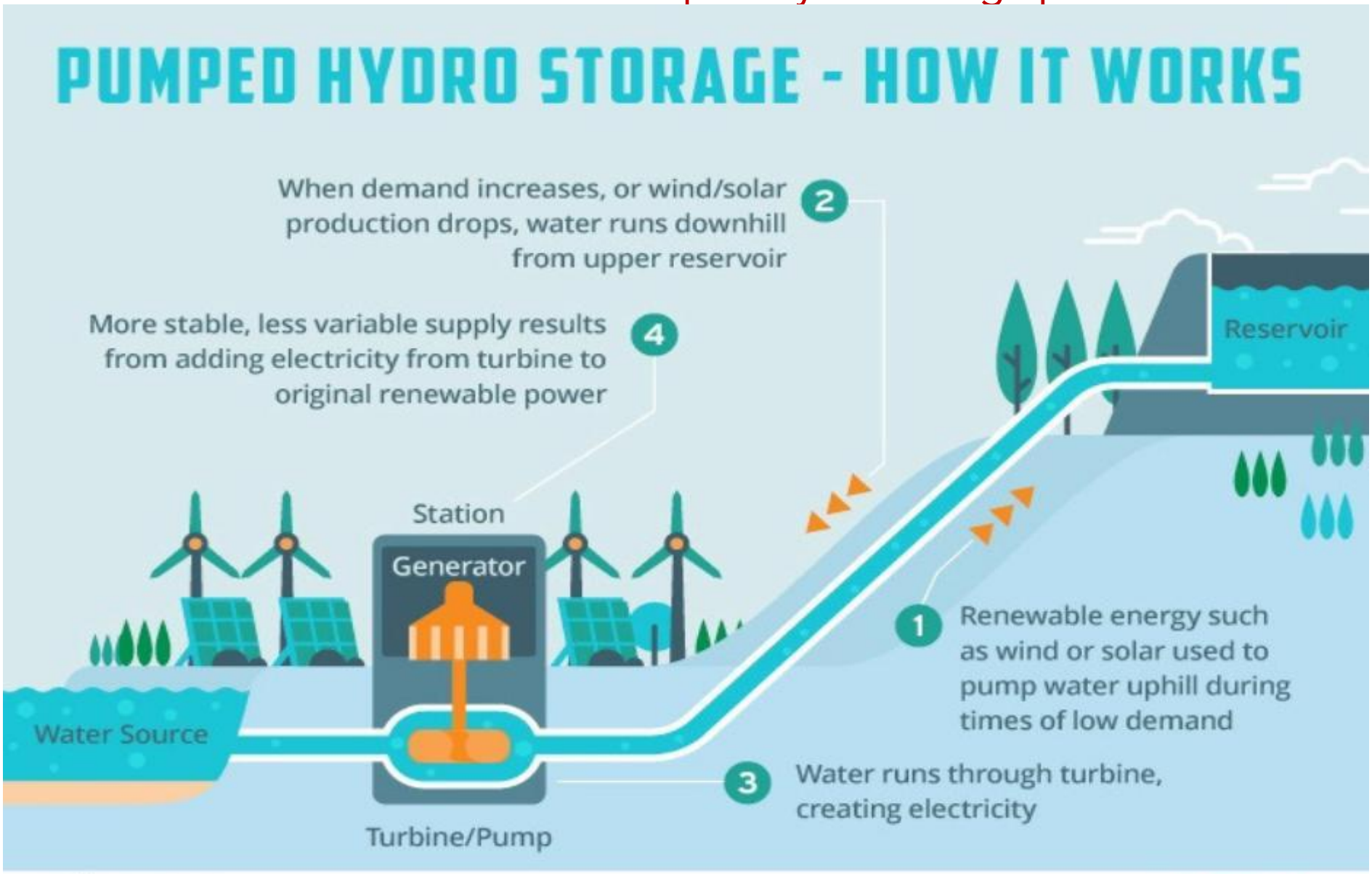


Schematic diagram of energy storage technologies

Ref: J. Mitali, S. Dhinakaran, A.A. Mohamad, Energy storage systems: a review, Energy Storage and Saving, Volume 1, Issue 3, 2022.

Mechanical Energy Storage Technologies

Pumped Hydro storage plant



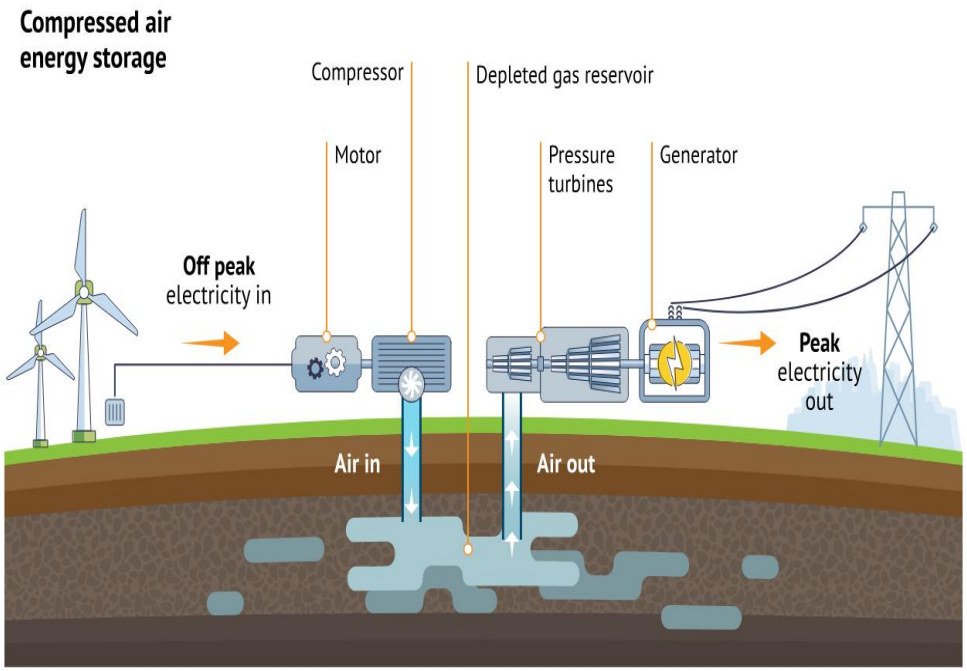
Disadvantages of Pumped Hydro Storage Plants (PHSP)

- **High Initial Investment**
- **Geographical Constraints**: Suitable sites require
- **Environmental Impacts:** Flooding of forests and agricultural land.Loss of biodiversity and wildlife habitats. Disturbance of river ecosystems. Changes in natural water flow and sediment transport.
- Water Loss
- Long Construction Time Construction may take 5–10 years or longer.

Future Directions Towards Sustainable Development

1. Closed-loop pumped hydro storage systems
2. Integration with solar and wind energy
3. Smart grid integration with AI-based energy management
4. Variable-speed pump-turbine technology
5. Environmentally sustainable reservoir design and water conservation
6. Climate-resilient infrastructure and net-zero energy storage solutions

Compressed Air Energy Storage (CAES) – Working Process (Highlight Points)



Excess / off-peak electricity availability

Surplus power from grid or renewables is used
Electric motor drives an **air compressor**

Air compression stage

Ambient air is compressed to high pressure (typically 40–70 bar)
Compression generates heat

Air storage

Compressed air is stored in **underground caverns**, **salt domes**, **aquifers**, or **high-pressure tanks**
Thermal energy may be stored separately (advanced CAES)

Energy storage phase

Energy is stored as **pressure energy of compressed air**
No electricity generation during this phase

Discharge / peak demand period

Stored compressed air is released in a controlled manner
Air is **heated** (using fuel or stored thermal energy)

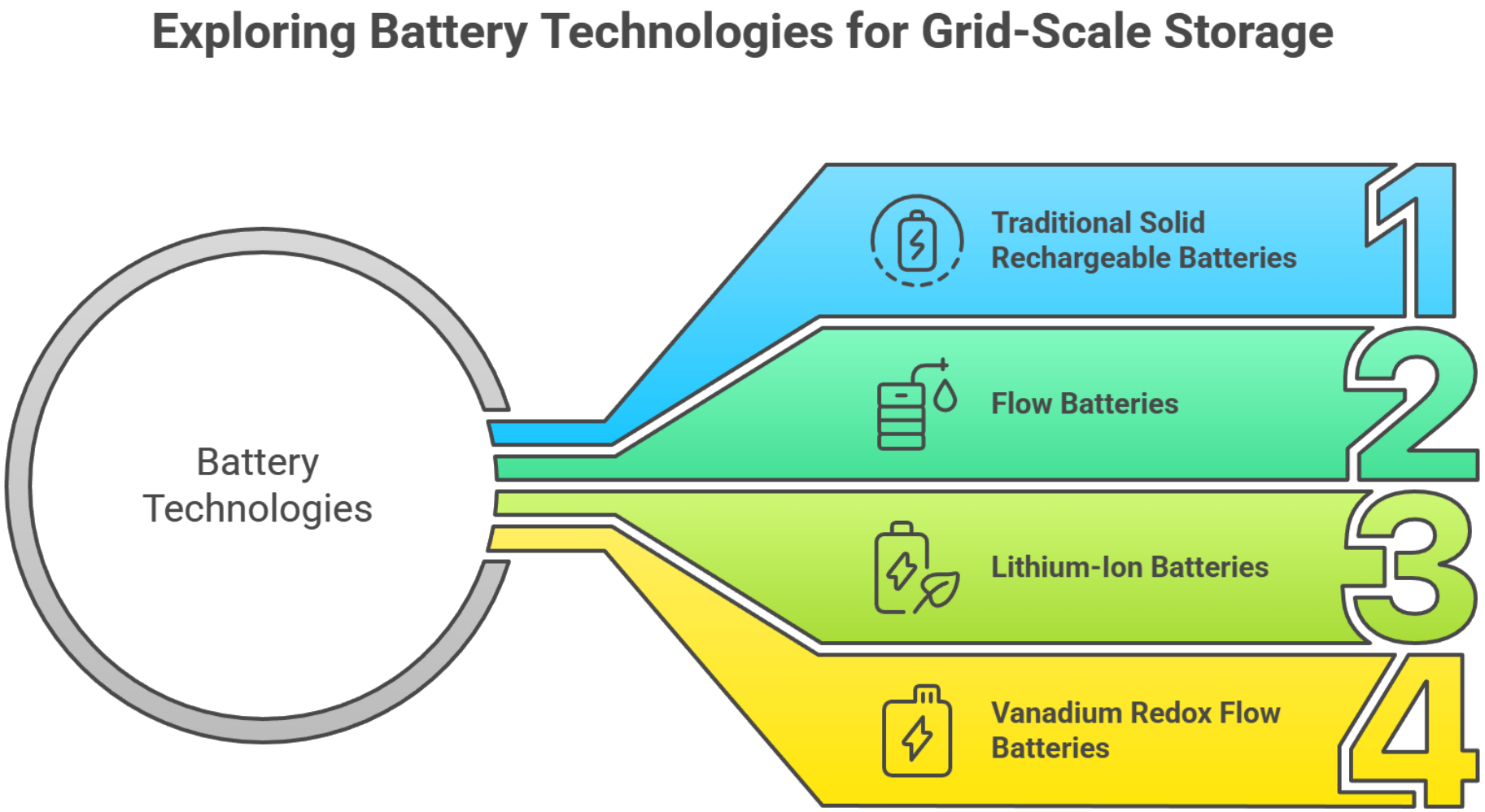
Power generation

Heated compressed air expands through a **gas turbine / expander**
Turbine drives a generator to produce electricity

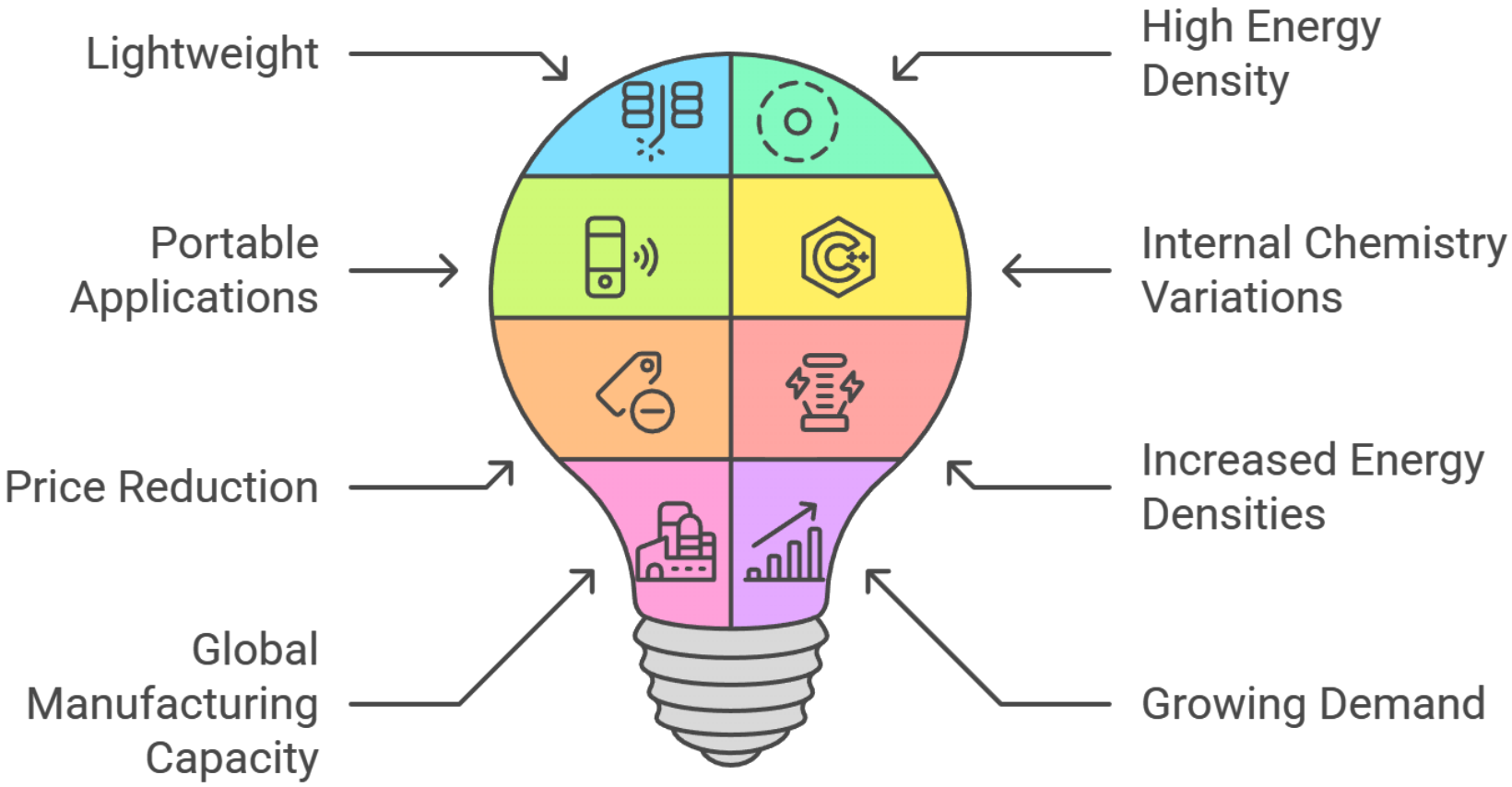
Electricity supply to grid

Generated power is fed back during peak demand
Fast ramp-up improves grid stability

Electrochemical Storage

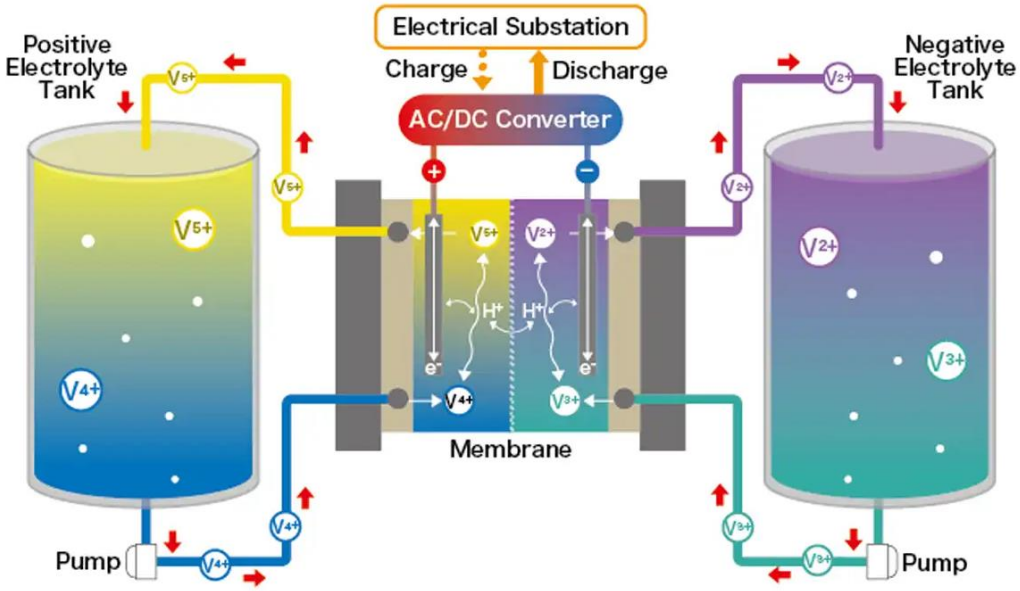


Lithium Ion Battery Overview



Flow batteries

- **Vanadium redox flow batteries (VRFBs)** operate through the **reduction and oxidation (redox)** reactions of vanadium ions in different oxidation states.
- **Vanadium** is an abundant and readily available material that can be obtained from natural mineral deposits as well as recovered from industrial waste streams.
- The battery consists of **two external electrolyte tanks** containing electrically conductive liquid electrolytes.
- One tank stores **positively charged vanadium ions (V^{5+})**, while the other contains **vanadium ions in the V^{2+} state**.
- The electrolytes are continuously circulated by **pumps** from the storage tanks to the battery's electrochemical cell (stack).
- Inside the cell, the vanadium ions undergo **redox reactions** across a membrane, allowing electrons to flow through an external circuit.
- This flow of electrons generates **electrical energy** that can be supplied to external loads.
- After the reaction, the electrolytes return to their respective tanks, where they can be reused, making the process **rechargeable and highly efficient**.
- Since the energy is stored in external electrolyte tanks, the **energy storage capacity** can be increased simply by using larger tanks, independent of the battery cell size.



•Charging:

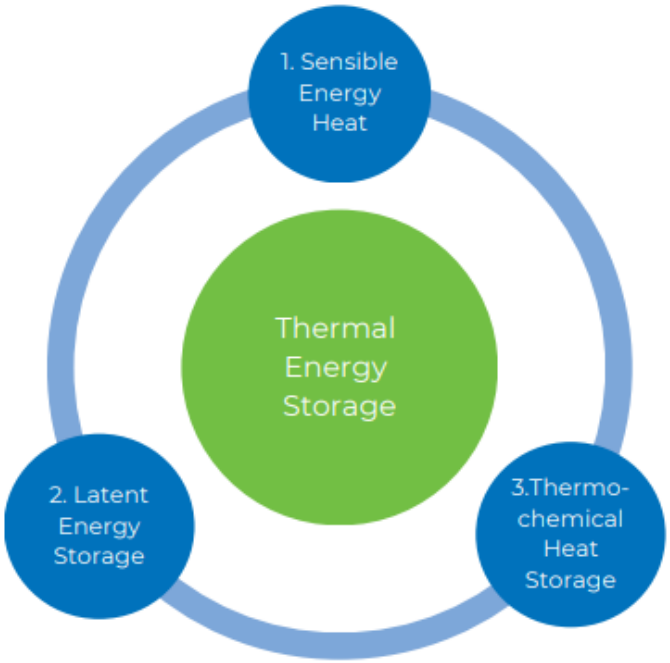
- *Positive Side:* $V^{4+} + H_2O \rightarrow VO^{2+} + 2H^+ + e^-$ (V^{4+} becomes V^{5+}/VO^{2+}).
- *Negative Side:* $V^{3+} + e^- \rightarrow V^{2+}$ (V^{3+} becomes V^{2+}).

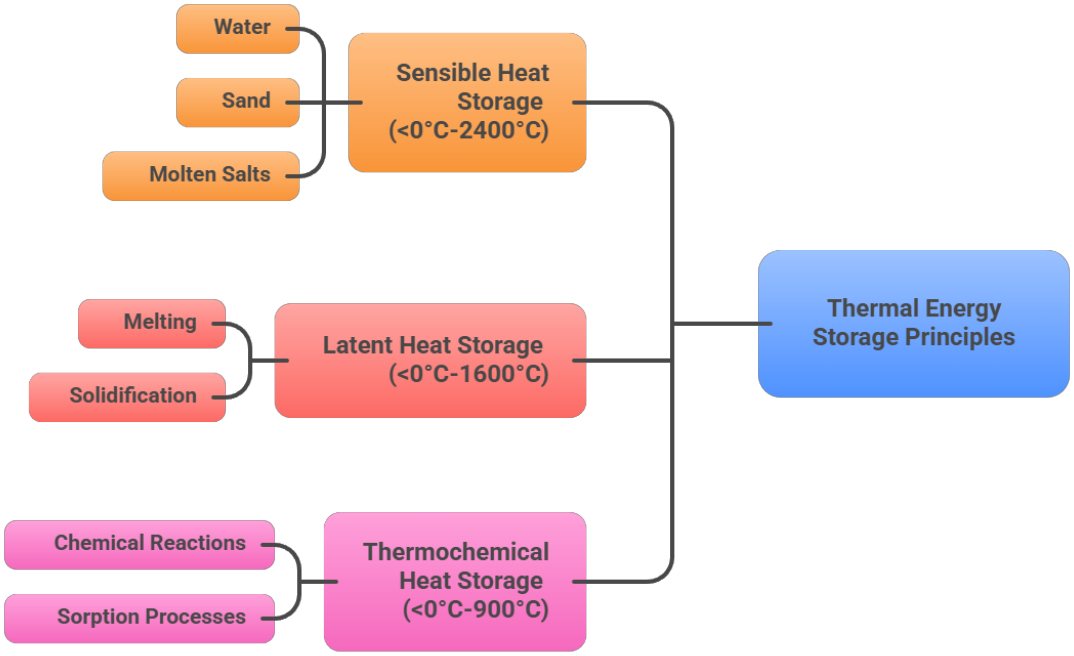
•Discharging (Reversed):

- *Negative Side:* $V^{2+} \rightarrow V^{3+} + e^-$ (V^{2+} becomes V^{3+}).
- *Positive Side:* $VO^{2+} + 2H^+ + e^- \rightarrow V^{4+} + H_2O$ (V^{5+}/VO^{2+} becomes V^{4+}).

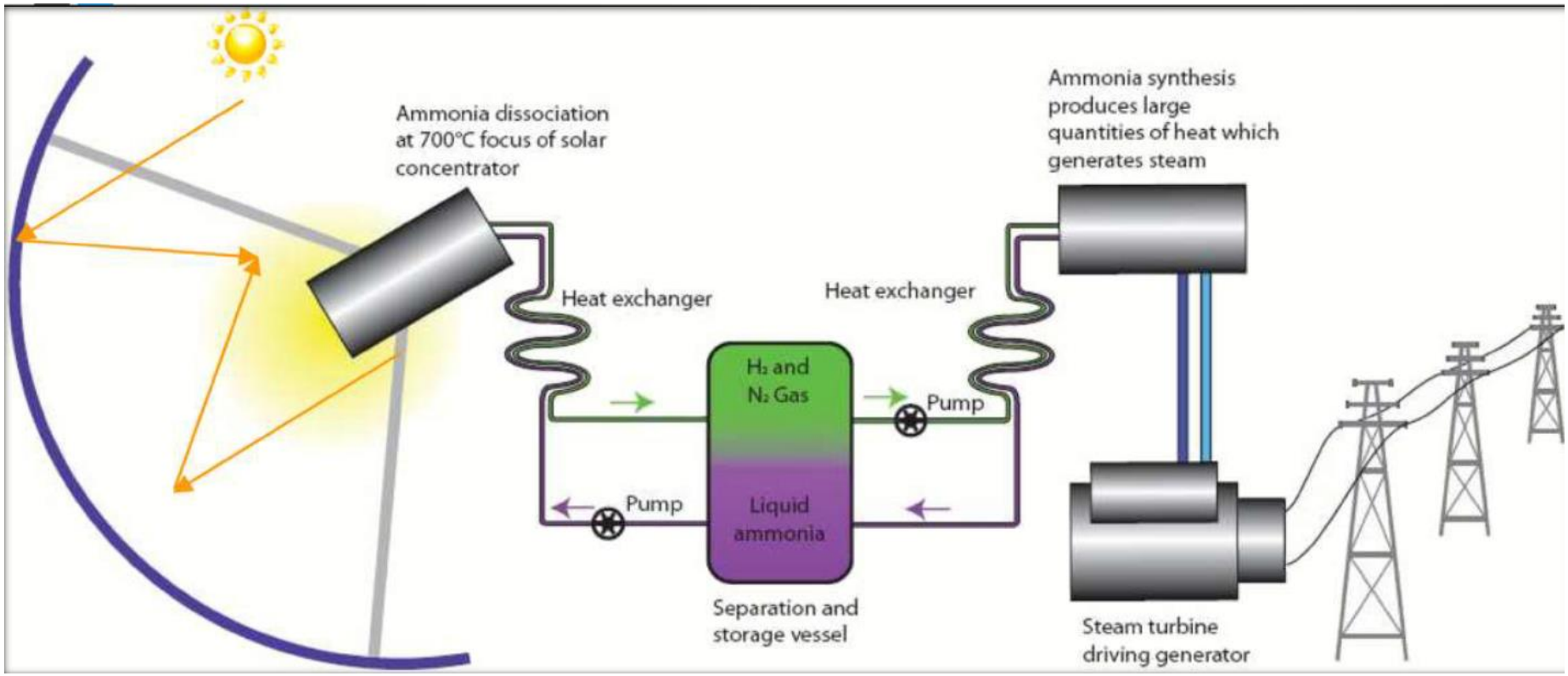
Image source: : https://www.huntkeyenergystorage.com

Thermal Energy Storage Technologies





AN AMMONIA BASED THERMO-CHEMICAL ENERGY STORAGE SYSTEM

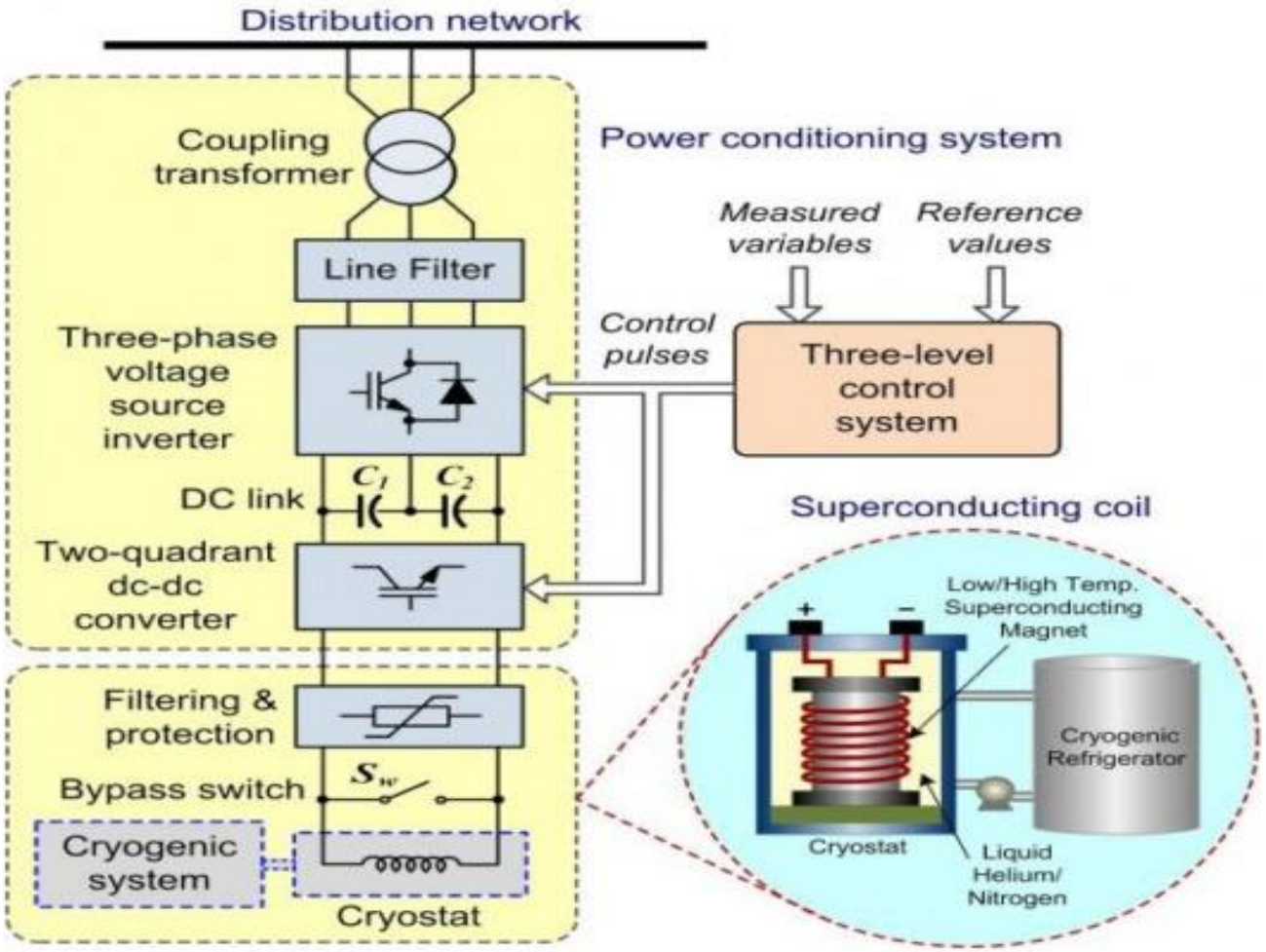


$$2NH_3 + \Delta H \longleftrightarrow N_2 + 3H_2$$

Electrical Energy Storage Technologies

Superconducting Magnetic Energy Storage

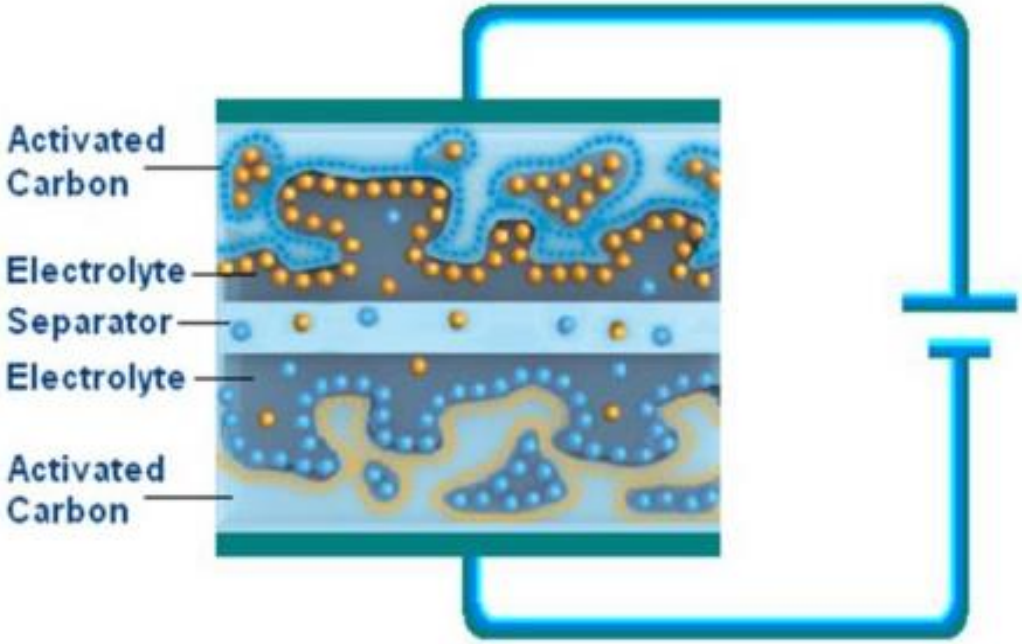
- A typical SMES system includes three parts: superconducting coil, power conditioning system and cryogenically cooled refrigerator.
- very small-time response (<<1 s) with high power peaks can be obtained.
- In addition, this device provides high efficiency, since the superconducting coil has virtually no Joule losses.



Super Capacitors

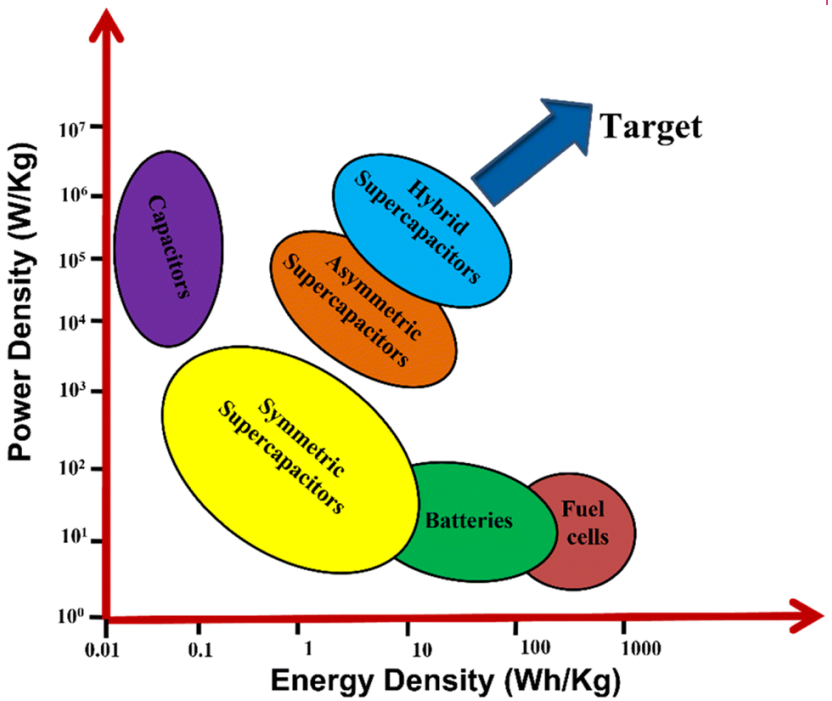
- **Activated Carbon:**
  - Extremely porous with a very large surface area.
  - Surface resembles a sponge.
  - Area allows more electrons to be stored than other conductors.
- **Principle**
  - Two layers consisting of nonporous electrodes
  - Separator is impregnated with an organic electrolyte
  - Thin separator can only withstand low voltages





Why Supercapacitors?

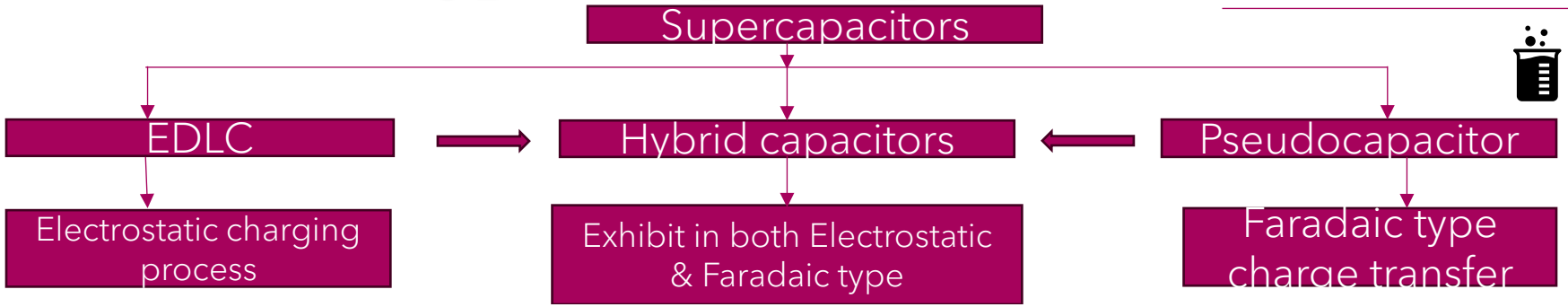
- High energy density & power density
- High capacitance (1mf – 10,000 F)
- Fast charging ability
- Low temperature performance
- Long cycle life (about 10 to 15 years compare to 5-10 years of li-ion battery)
- Eco-friendly

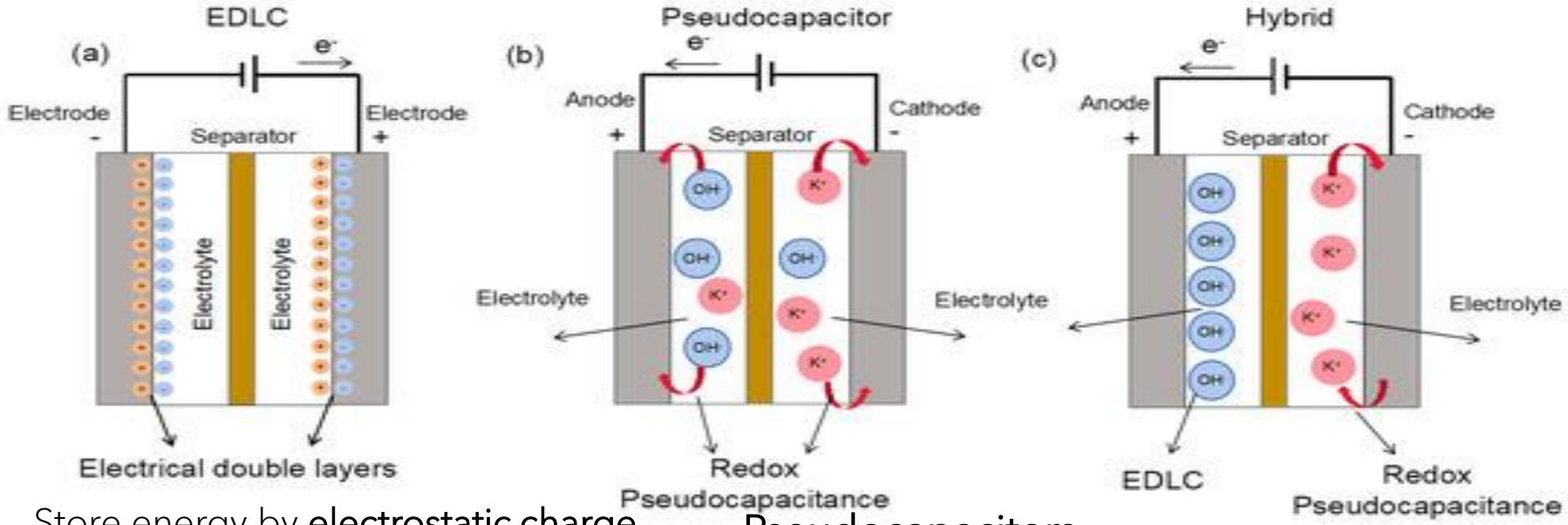


Ragon plot of energy storage devices

Ref: Amtul Nashim, Ritik Mohanty, Priyadarshi K. Ray and K. M. Parida. RSC advances. 13, 2023.

Types Supercapacitors





1. Store energy by **electrostatic charge separation** at the electrode–electrolyte interface.
2. Electrode materials: Activated carbon, graphene, carbon nanotubes.

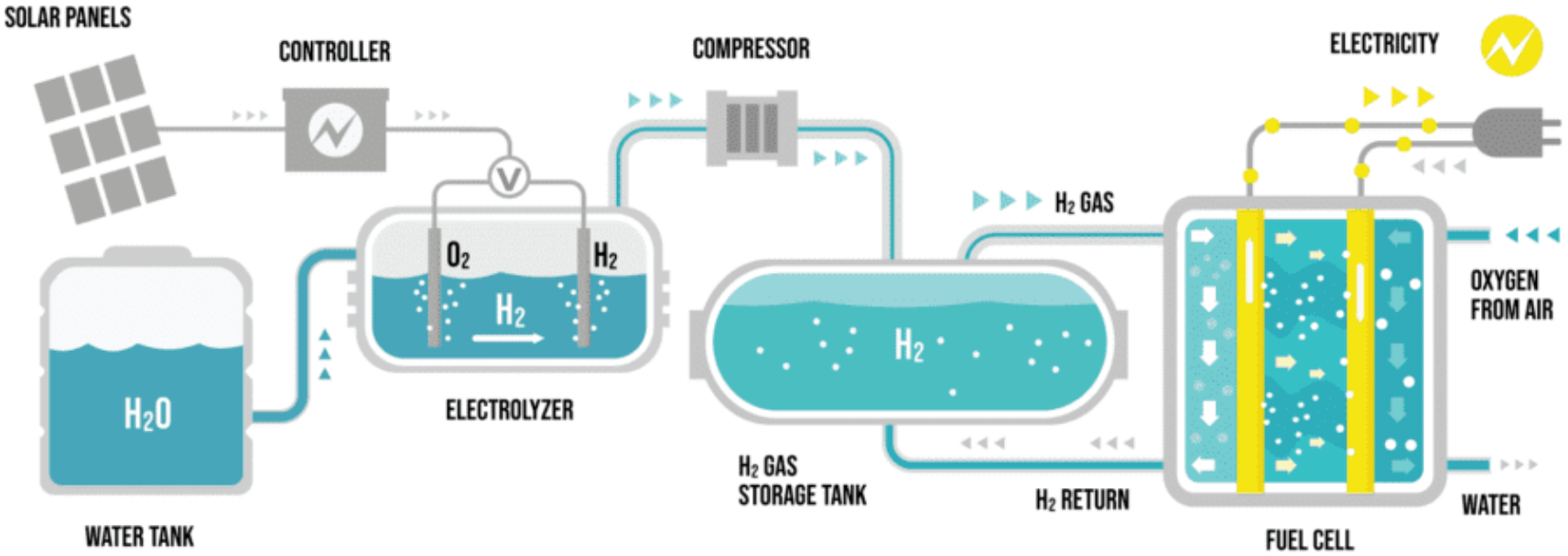
Pseudocapacitors

1. Store energy through **fast reversible redox (Faradaic) reactions** on the electrode surface.
2. Electrode materials: Metal oxides (RuO_2, MnO_2) and conducting polymers (PANI, PPy).

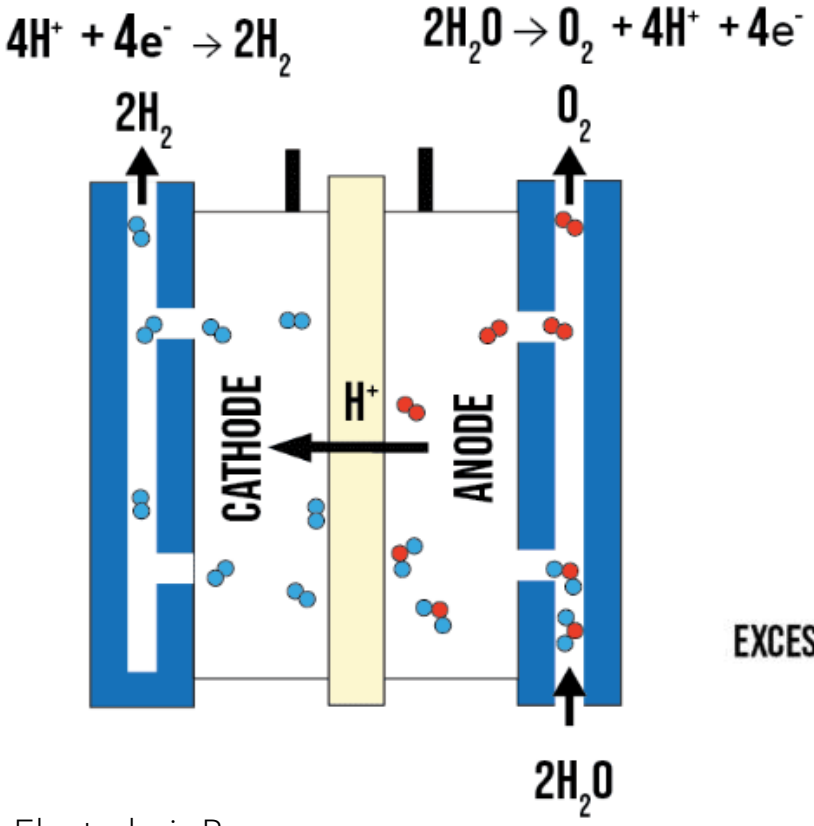
Chemical Energy storage technologies-

Fuel Cells and Electrolysers

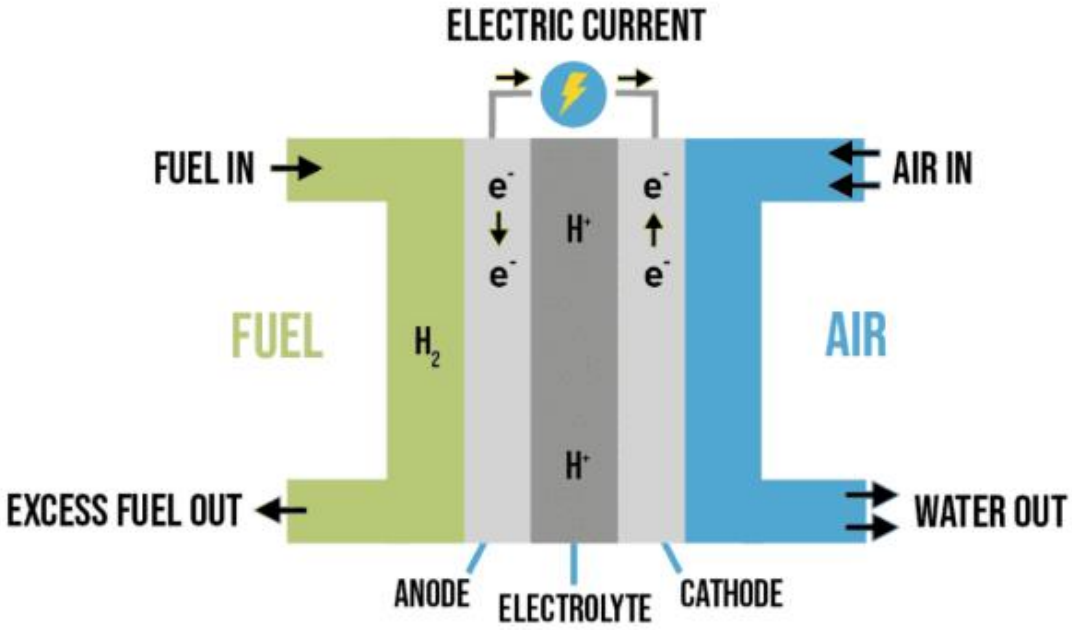
- Electrolyzers and fuel cells are critical components of green energy systems as they enable the efficient storage and utilization of renewable energy. **Electrolyzers allow for the conversion of excess renewable energy**, such as wind or solar power, **into hydrogen gas** through electrolysis. This hydrogen gas can then be stored for later use or used directly in fuel cells to generate electricity.
- On the other hand, fuel cells produce electricity through a chemical reaction between hydrogen and oxygen, emitting only water vapor as a byproduct. By utilizing renewable energy sources and producing no harmful emissions, **fuel cells and electrolyzers play a vital role in reducing greenhouse gas emissions and combating climate change**.



Electrolysis and Fuel Cell process



Electrolysis Process



Fuel Cell Process

$2H_2O$ + Electrical Energy → $2H_2$ + O_2

Image source: Totalshield.com

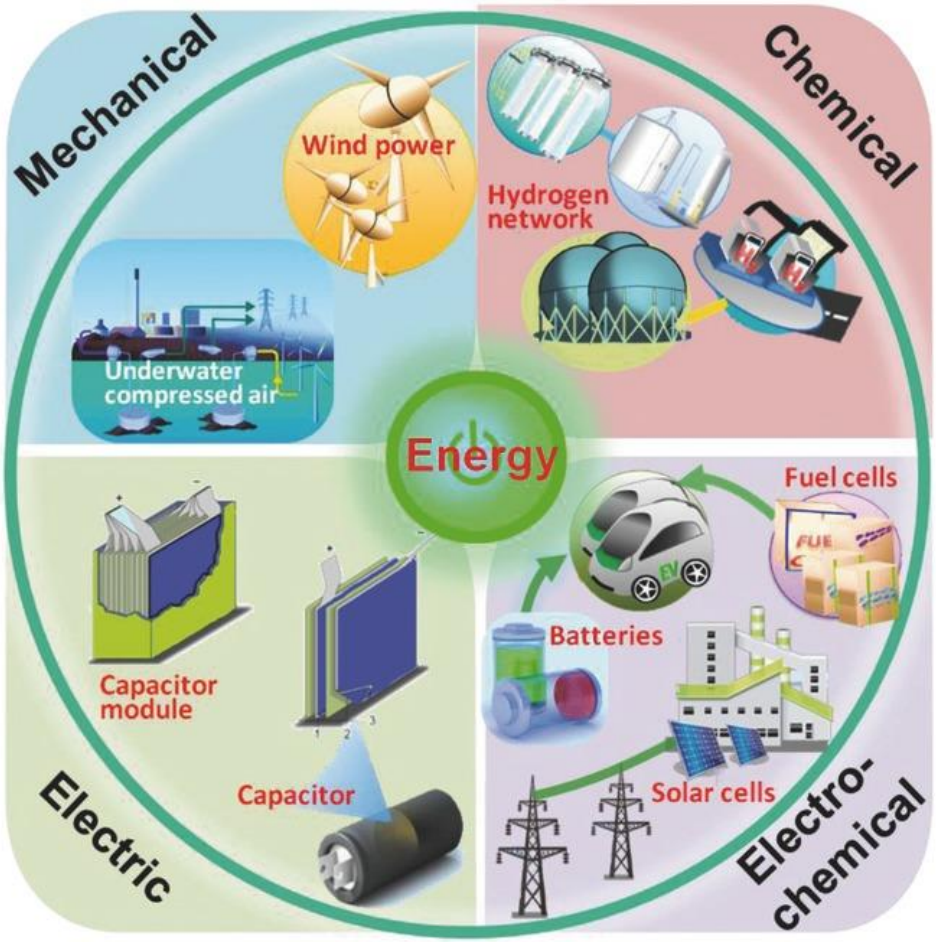
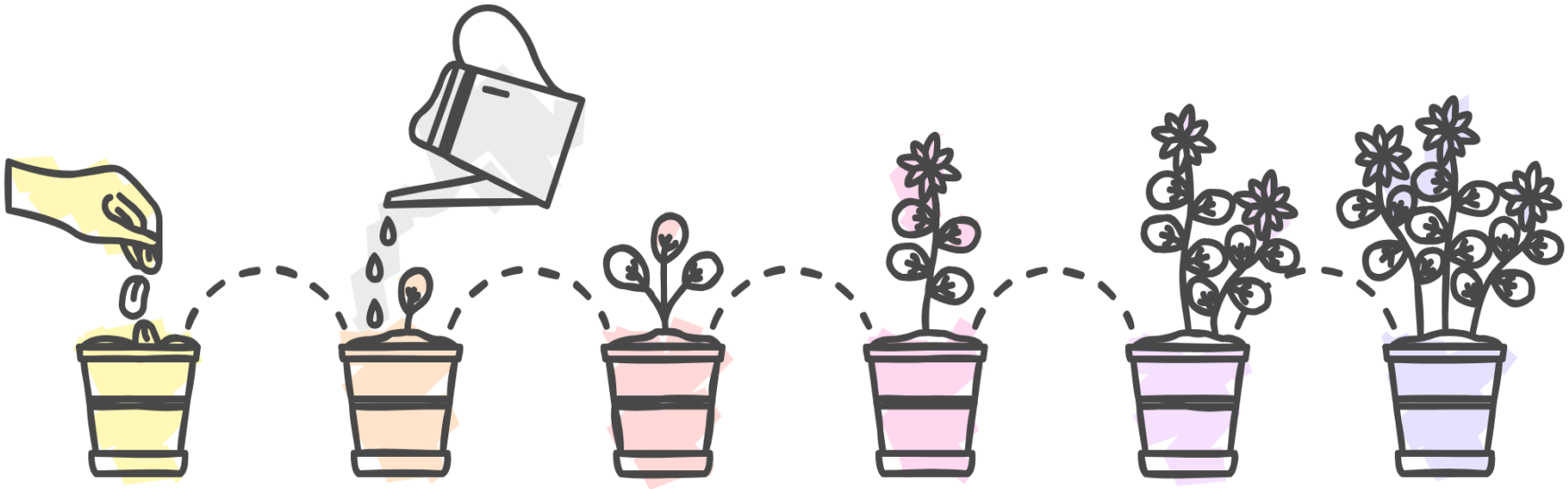


Figure (2) Energy storage systems

Ref: https://www.linquip.com/blog/types-of-energy-storage-methods/

India's Climate Action Plan to the United Nations Framework Convention on Climate Change (UNFCCC)



Current State

High carbon emissions and reliance on fossil fuels

Non-Fossil Energy

Reach 500GW non-fossil energy capacity by 2030

Renewable Energy

50% energy requirements from renewables by 2030

Carbon Emission Reduction

Reduce total projected carbon emissions by 1 billion tonnes by 2030

Carbon Intensity Reduction

Reduce carbon intensity of economy by 45% by 2030

Net Zero Emissions

Achieve net zero emissions by 2070

Sustainable Agriculture

Sustainable Agriculture

Sustainable agriculture is the practice of producing food in ways that protect the environment, conserve natural resources, support farmers' livelihoods, and ensure food production for future generations.

Key Elements of Sustainable Agriculture

- **Environmental sustainability:** Conserves soil, water, biodiversity, and reduces pollution.
- **Economic sustainability:** Ensures farming is profitable and economically viable.
- **Social sustainability:** Improves the quality of life for farmers, workers, and rural communities.
- **Resource efficiency:** Uses land, water, energy, and other inputs efficiently.
- **Climate resilience:** Helps agriculture adapt to and mitigate climate change.

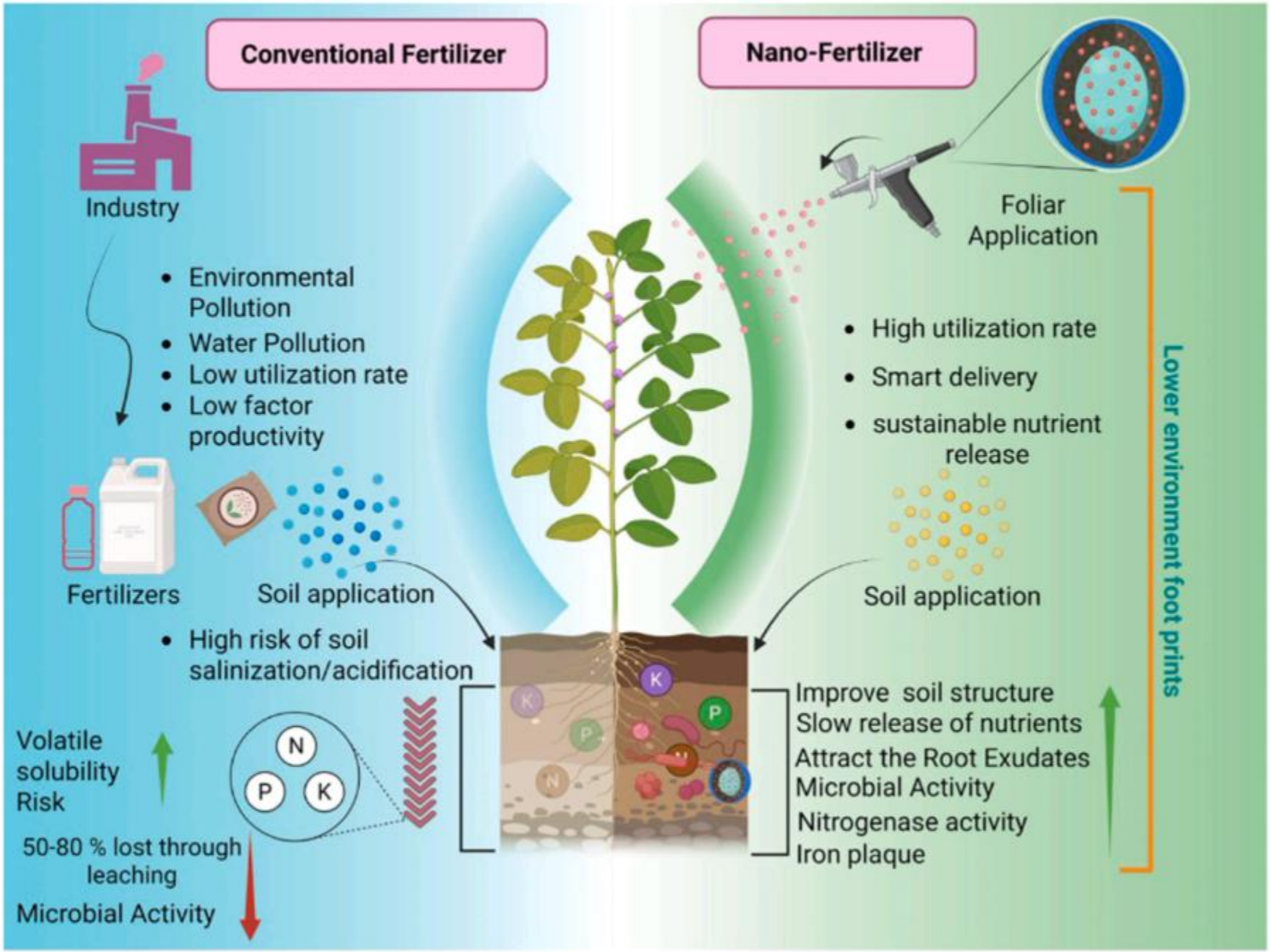
Sustainable agriculture integrates three main goals

- ***Environmental health***
- ***Economic profitability***
- ***Social and economic equit***

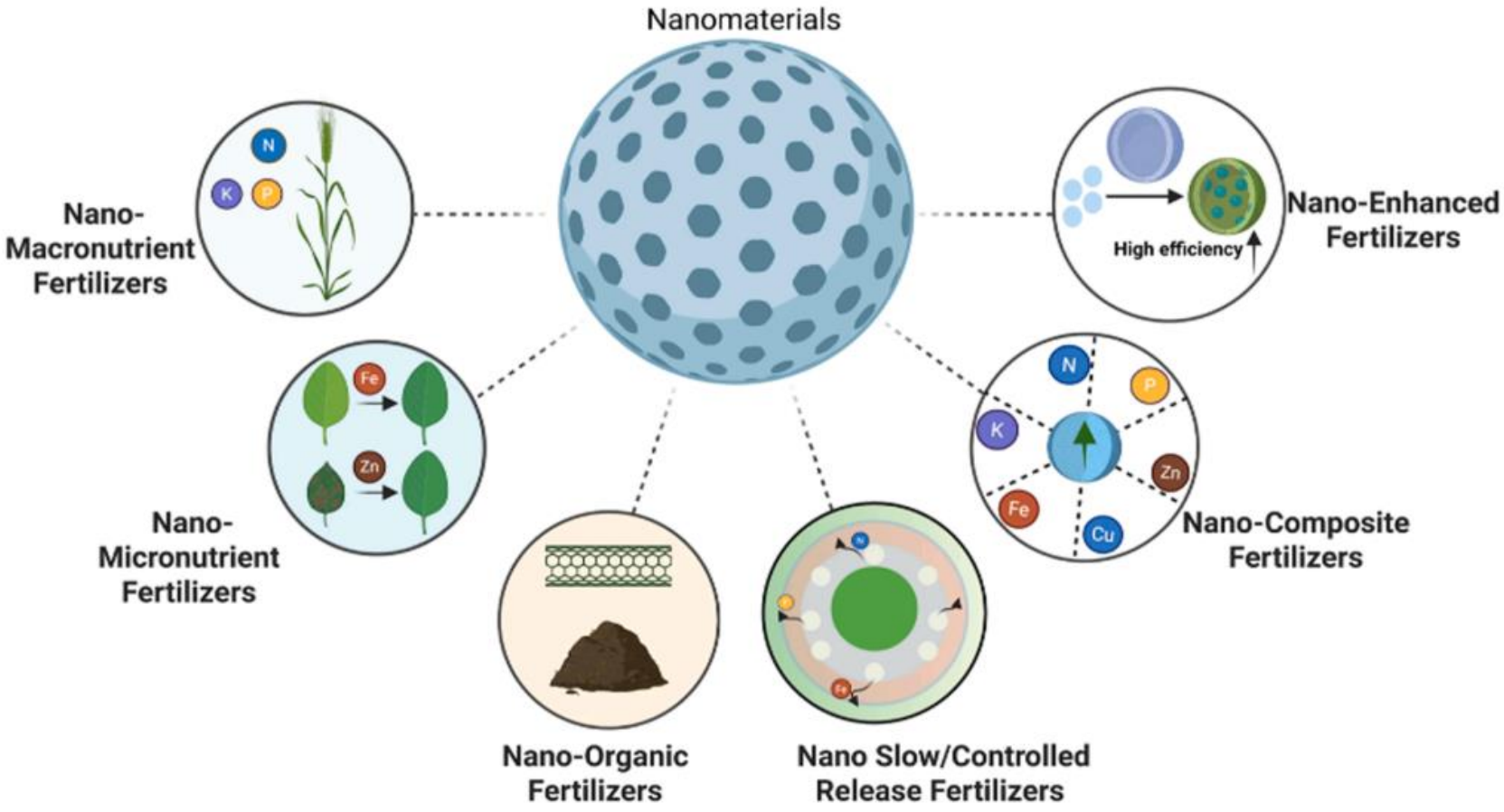


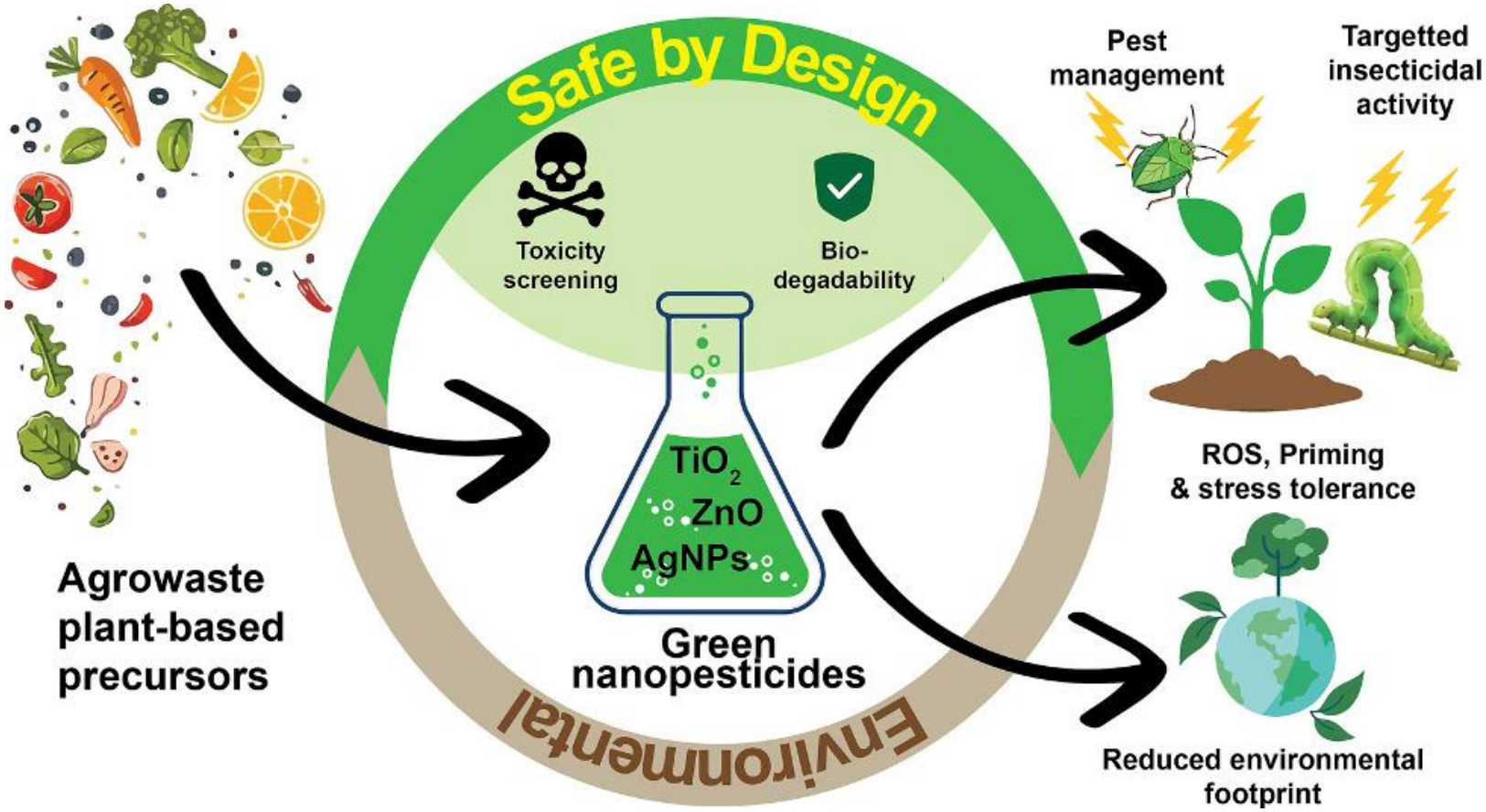


Fig. 1. Variables involved in agricultural sustainability assessment


Conventional Fertilizer
Nano-Fertilizer
Industry
Foliar Application
Environmental Pollution
Water Pollution
Low utilization rate
Low factor productivity
High utilization rate
Smart delivery
sustainable nutrient release
Fertilizers
Soil application
Soil application
High risk of soil salinization/acidification
Lower environment foot prints
Improve soil structure
Slow release of nutrients
Attract the Root Exudates
Microbial Activity
Nitrogenase activity
Iron plaque
Volatile solubility Risk
50-80 % lost through leaching
Microbial Activity
N
P
K

Nanofertilizer




Safe by Design
Toxicity screening
Bio-degadability
Agrowaste plant-based precursors
TiO2
ZnO
AgNPs
Green nanopesticides
Environmental
Pest management
Targetted insecticidal activity
ROS, Priming & stress tolerance
Reduced environmental footprint

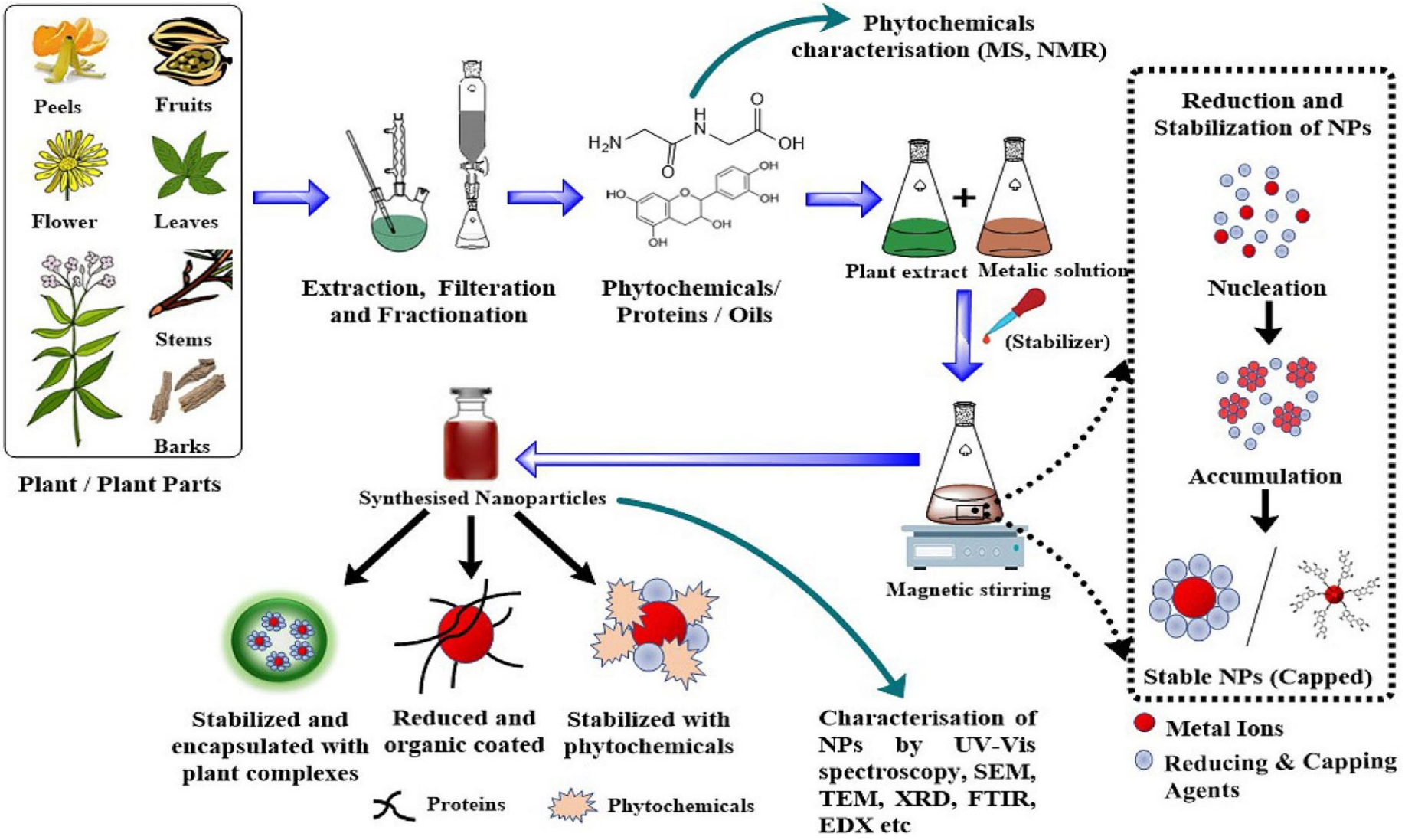
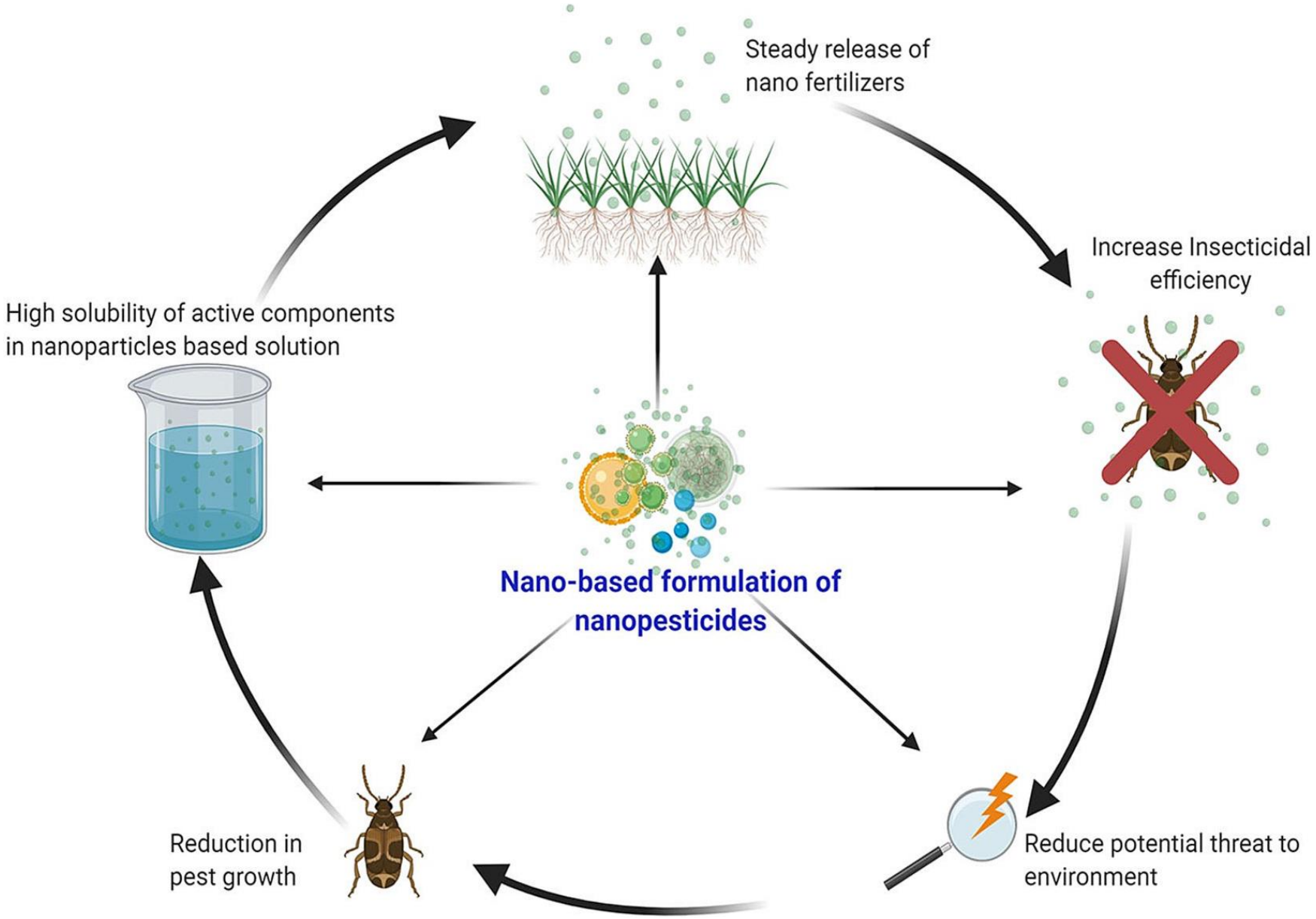


Fig. 1. Overview of green synthesis strategies for nanopesticide production using plant-derived bioreductants. The schematic illustrates the primary biosynthetic routes involving fruit peels, leaves, and root extracts rich in phytochemicals (e.g., polyphenols, flavonoids, alkaloids), which act as both reducing and capping agents. This approach allows nanoparticle formation under mild conditions without external chemical additives.


Steady release of
nano fertilizers
Increase Insecticidal
efficiency
High solubility of active components
in nanoparticles based solution
Nano-based formulation of
nanopesticides
Reduction in
pest growth
Reduce potential threat to
environment

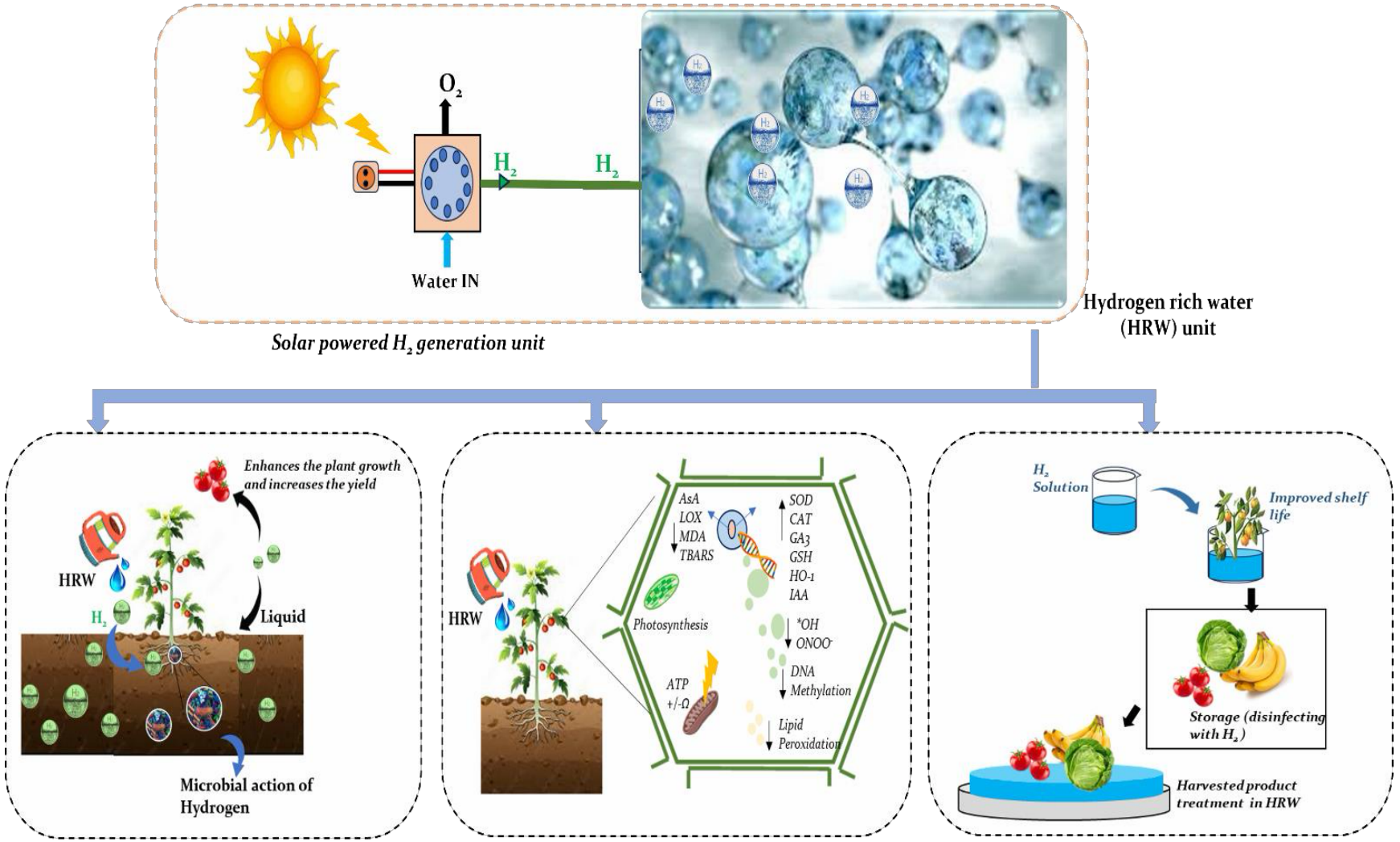


Fig. 2 Schematic representation of hydrogen based Agronomy

Importance of Sustainable Development

Importance of Sustainable Development

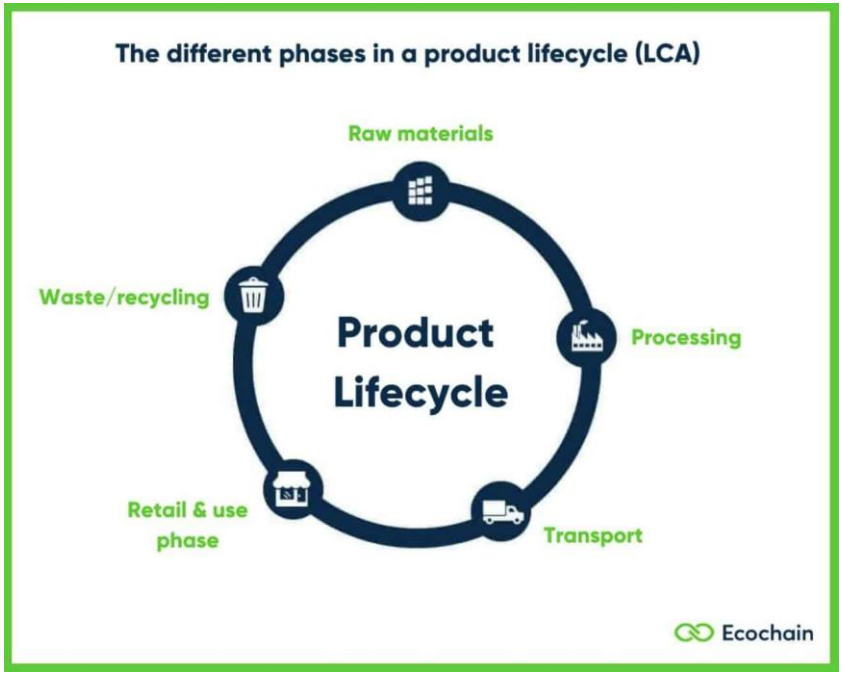
- **Meets Basic Human Needs:** Ensures long-term availability of food, water, shelter, and energy.
- **Supports Sustainable Agriculture:** Promotes efficient farming practices while protecting soil and natural resources.
- **Combats Climate Change:** Reduces greenhouse gas emissions by encouraging renewable energy use.
- **Promotes Economic Stability:** Creates sustainable jobs and supports long-term economic growth.
- **Protects Biodiversity:** Conserves ecosystems and maintains the balance of plant and animal life.
- **Conserves Natural Resources:** Encourages responsible use of resources for future generations.
- **Improves Quality of Life:** Balances economic development, environmental protection, and social well-being.

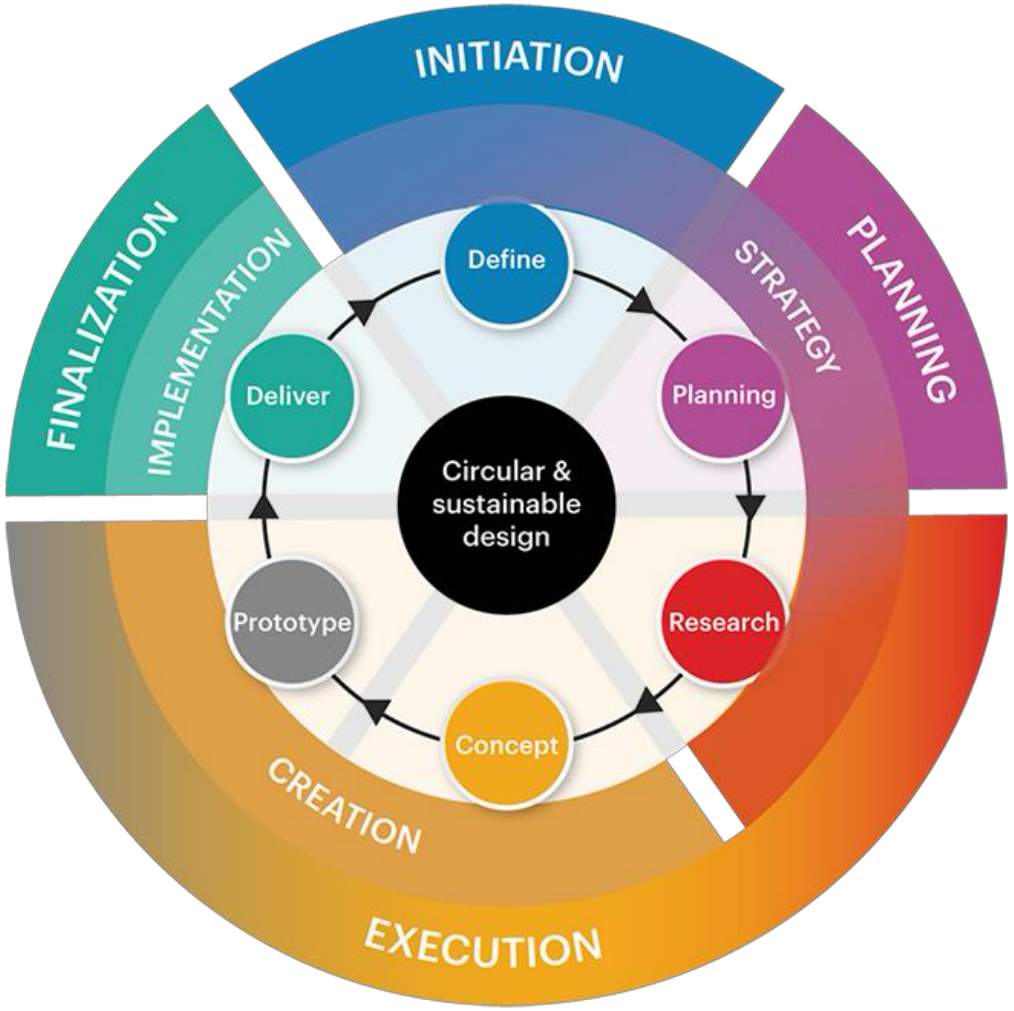
Why Sustainability in Design?

- **Reduces environmental impact**
- **Enhances brand reputation**
- **Encourages innovation**
- **Meets regulatory requirements**
- **Saves costs over product lifecycle**

Life-Cycle Thinking

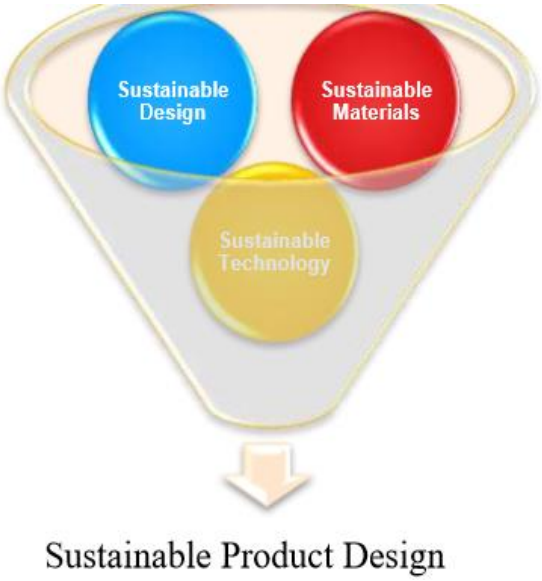
- Assess product impact from raw material extraction to disposal
- Identify improvement opportunities **at each stage**
- Apply Life Cycle Assessment (LCA) tools


The different phases in a product lifecycle (LCA)
Raw materials
Processing
Transport
Retail & use phase
Waste/recycling
Product Lifecycle
Ecochain


INITIATION
PLANNING
EXECUTION
CREATION
IMPLEMENTATION
FINALIZATION
STRATEGY
Define
Planning
Research
Concept
Prototype
Deliver
Circular & sustainable design

Sustainable Product Design and Development Techniques

Sustainable Product Design



How to make a sustainable product design?

- Make it less complex
- Make it more useful
- Reduce Material Variety
- Avoid toxic or harmful materials and chemicals
- Reduce size and weight
- Design for Upgradeability
- Create durable and high quality designs
- Design for life after death

PRINCIPLES OF SUSTAINABLE PRODUCT DESIGN

- Efficient use of raw materials
- Resource efficient design
- Durable design
- High disposability
- Efficient logistics
- Service substitution
- Proper training and education for the workers

Strategies for Sustainable design development

- Design for longevity
- Design for disposal

1. Design for Longevity

- ✓ Timeless, classical design
- ✓ Long lifespan
- ✓ Durability
- ✓ Designed for ease of repair and maintenance
- ✓ Designed for modularity
- ✓ Possibilities for reuse
- ✓ Possibility to upgrade with latest technology

2. Design for Disposal

- ❖ Fashionable design
- ❖ Designed for takeback
- ❖ Designed for recycling
- ❖ Designed for eco-friendly disposal

What is Sustainable Manufacturing?

Manufacturing

Mechanical, physical and chemical process to modify the geometry, properties and appearance of a given starting material in the making of new finished part of product.

Sustainable manufacturing

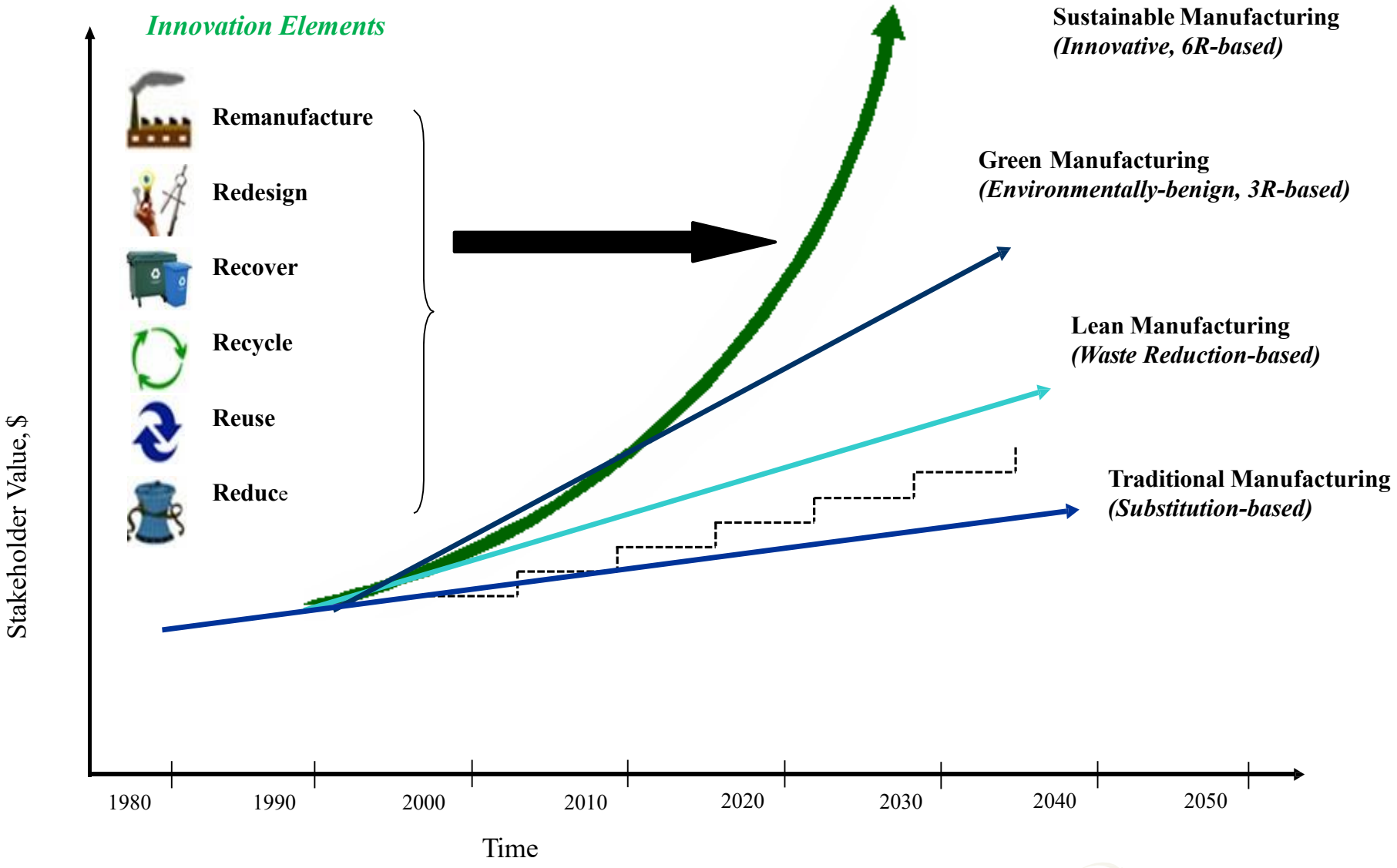
The creation of manufactured products that use processes that are **non-polluting, conserve energy and natural resources, and are economically sound and safe for employees, communities, and consumers.**"

(The United States, Department of Commerce defined Sustainable manufacturing)

Basic Objective of sustainability in Manufacturing activity

- Minimise manufacturing cost
- Minimise energy consumption
- Reduce environmental impact
- Improve personnel health
- Improve operational safety
- Improve waste management practices

Evolution of Sustainable Manufacturing



Sustainable Manufacturing

Sustainability of a manufacturing process is to be defined, measured and evaluated using six sustainability criterion.



Sustainable Manufacturing in Practice

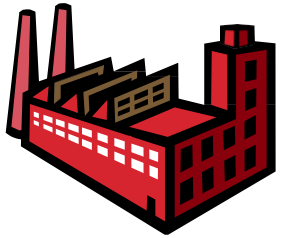
Using renewable materials that don't deplete the natural environment



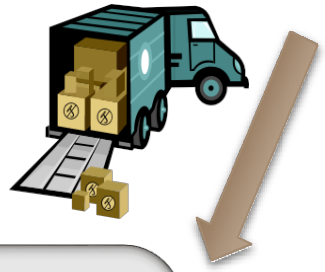
Using fewer materials and inputs and materials that are non-hazardous or recycled



Modifying production processes to use less energy, water, and materials and to produce less waste



Using less packaging, lowering product weight, using more efficient transportation and logistics



Working with stakeholders like customers and retailers to reduce the environmental impact of sales and distribution



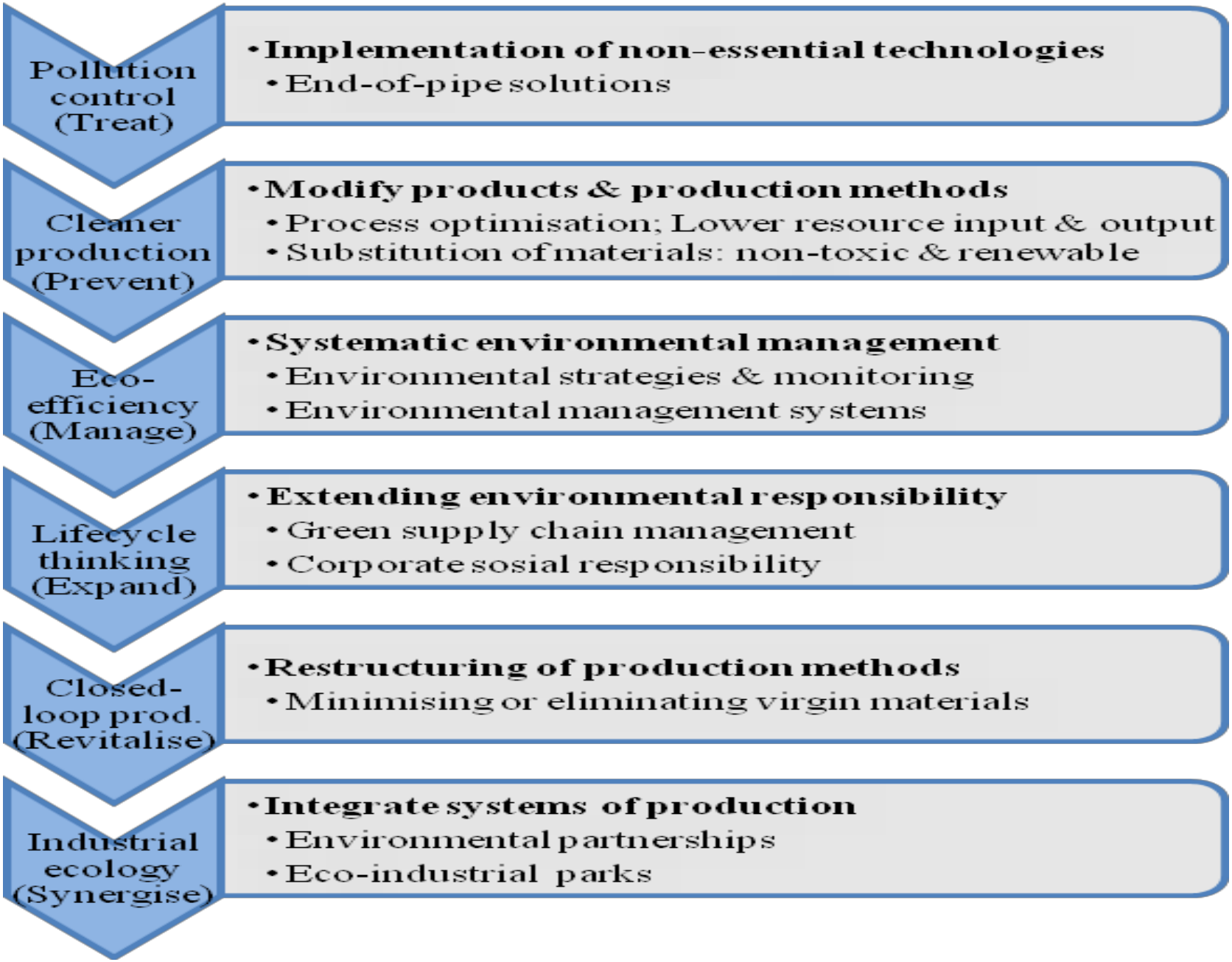
Expanding the life of the product, making it easier to repair or designing it to use fewer resources during use



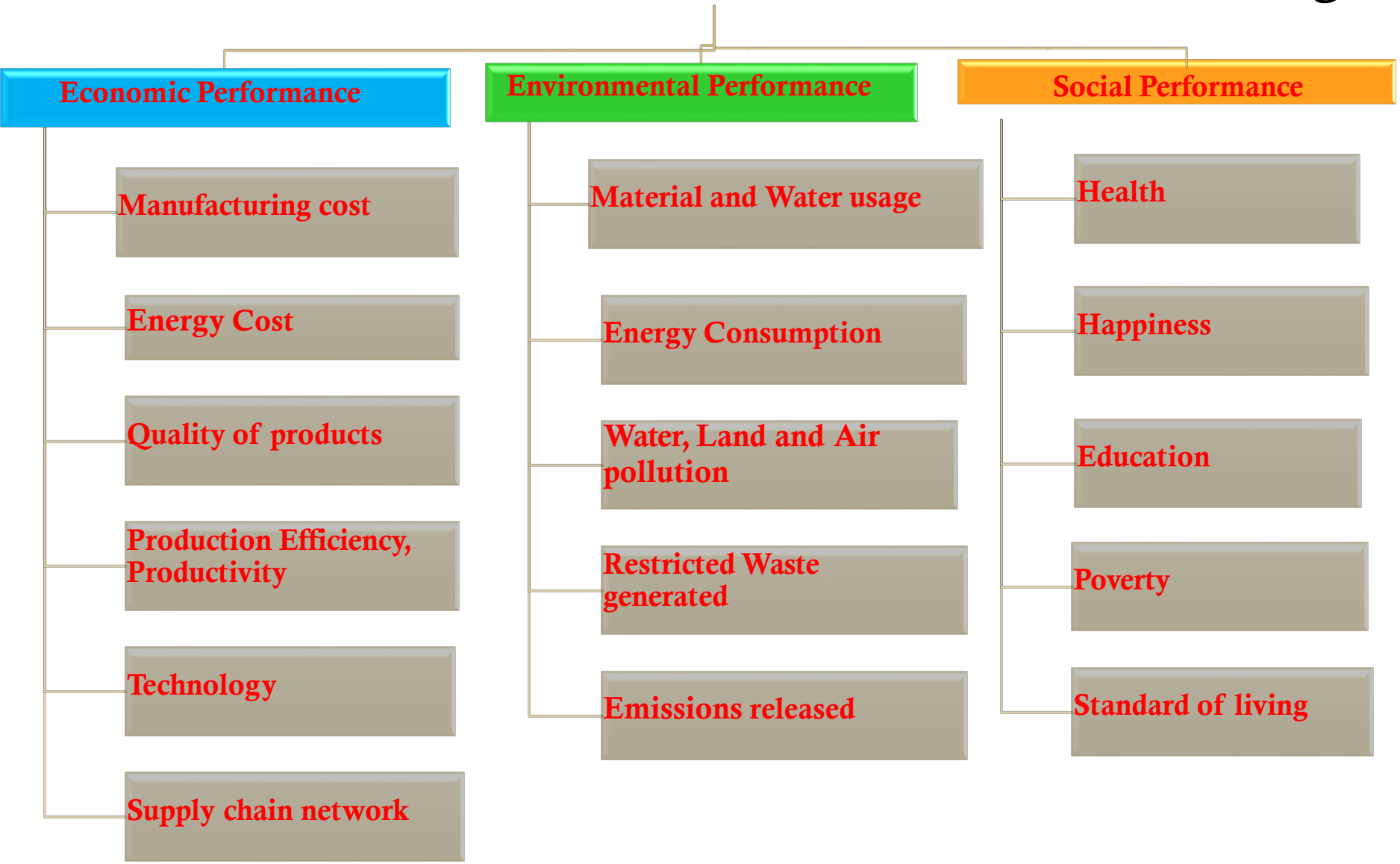
Designing your product to be reusable, remanufacturable, recyclable, or biodegradable

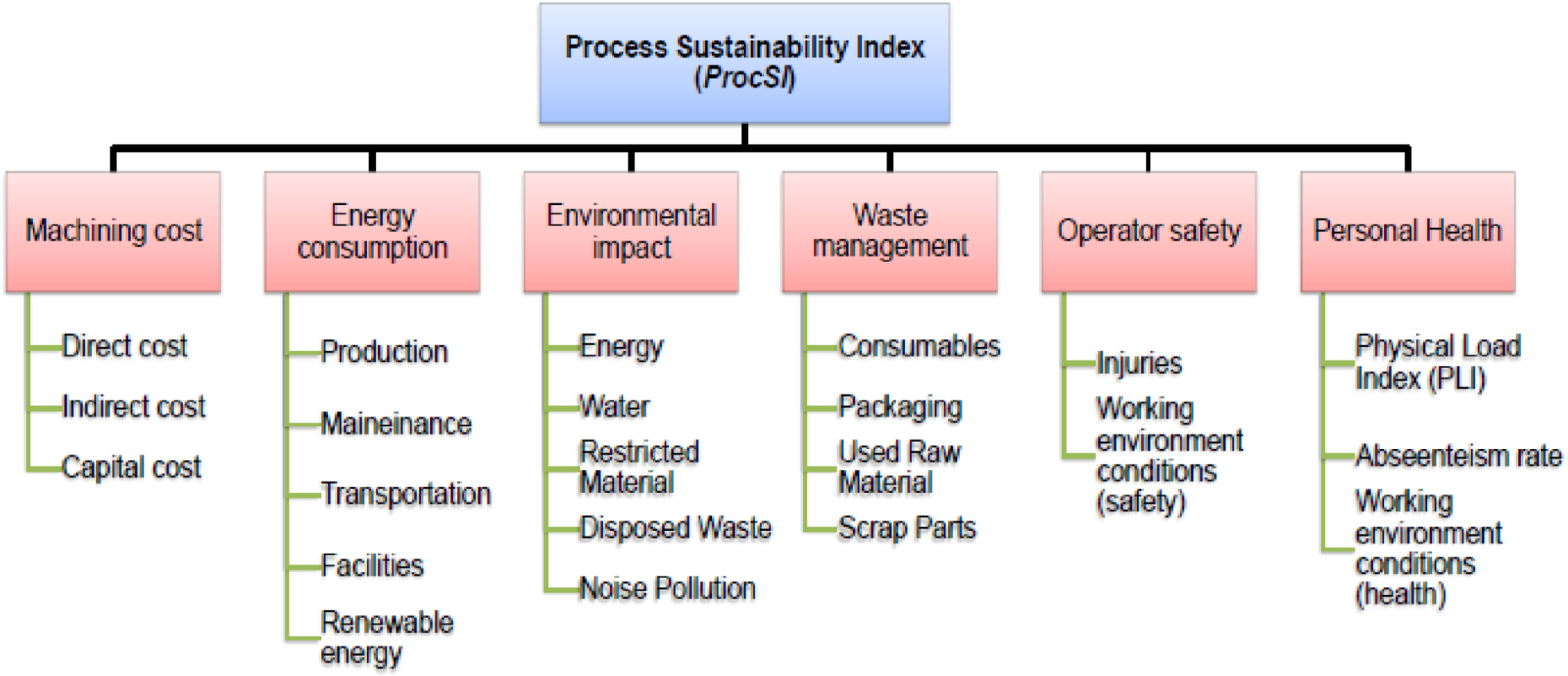


Sustainable manufacturing practices



Performance Indicators for Sustainable Manufacturing




Process Sustainability Index (ProcSI)
Machining cost
Direct cost
Indirect cost
Capital cost
Energy consumption
Production
Maineinance
Transportation
Facilities
Renewable energy
Environmental impact
Energy
Water
Restricted Material
Disposed Waste
Noise Pollution
Waste management
Consumables
Packaging
Used Raw Material
Scrap Parts
Operator safety
Injuries
Working environment conditions (safety)
Personal Health
Physical Load Index (PLI)
Abseenteism rate
Working environment conditions (health)

Indicators for Sustainability evaluation

Cost

- Labour Cost (Rs./unit)
- Cost of consumables (Rs./unit)
- Cost of energy (Rs./unit)
- Maintenance Cost (Rs./unit)
- Cost of by product treatment (Rs./unit)
- Indirect labour cost (Rs./unit)

Energy Consumption

- In line energy consumption (KWh/Unit)
- Energy consumption on maintaining facility environment (KWh/Unit)
- Energy consumption in transportation in/out of line (KWh/Unit)
- Ratio of use of renewable energy (%)

Environmental Impact

- GHG emissions from energy consumption from line (Ton Co2 Equivalent/ unit)
- Total water consumption (Kgs/unit)
- Mass of restricted waste generated (Kgs/unit)
- Noise level inside/ outside factory (dB)
- Ration of renewable energy used (%)

Indicators for Sustainability evaluation

Operator Safety

- Exposure to Corrosive/Toxic chemicals (points/ person)
- Exposure to high energy (points/person)
- Injury rate (Injuries/person)

Personnel Health

- Chemical Contamination of working environment (mg/m^3)
- Mist/Dust level (mg/m^3)
- Noise level (dB)
- Physical load index (dimensionless)
- Health related absenteeism rate (%)

Waste Management

- Mass of Disposed consumables (kg/unit)
- Consumables reuse ratio (%)
- Mass of mist generation (kg/unit)
- Mass of disposed chips and scraps (kg/unit)
- Ratio of recycled chips and scraps (%)

Tools for sustainability measurement and impact assessment

- Sustainability is measured by assessing performance of Social, Environmental, and Economic principles.
- Tools For Sustainability Measurement:
- For Environment: LCA - Life cycle assessment
- For Social: SLCA - Social life cycle assessment
- For Economic: LCC - Life cycle costing
- LCSA=LCA+SLCA+LCC
- where: LCSA - Life cycle sustainability assessment
- Life cycle sustainability assessment - refers to the evaluation of all environmental, social and economic negative impacts and benefits in decision-making processes towards more sustainable products throughout their life cycle.

THANK YOU